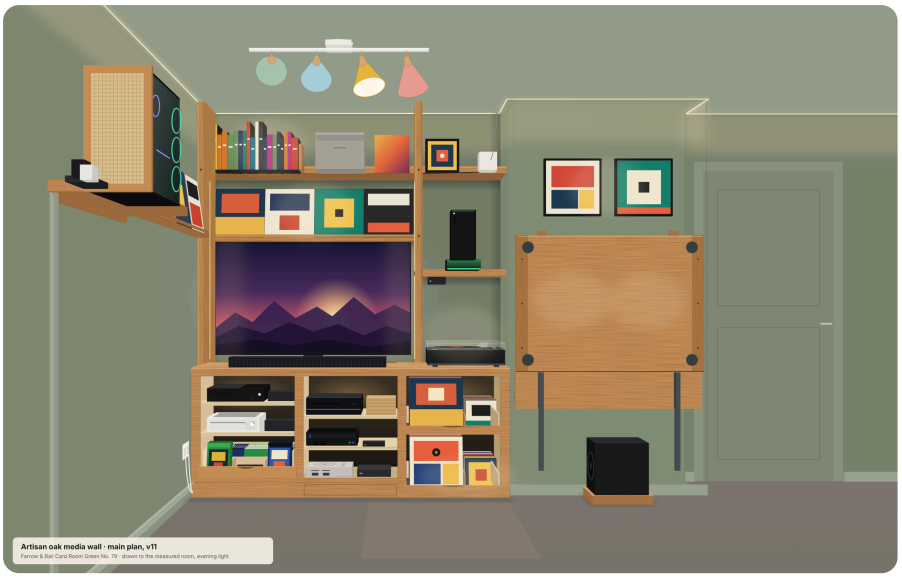
Home Network & Media Wall Plan

Revised Oct 18, 2026 · @emma · version 11, drawn to your measurements

What changed in v11. The TV moves **25.25 cm nearer the side wall**. The two narrow gaps beside it become one 50.5 cm zone, so the **turntable comes back into the alcove**, with the Series X on an oak shelf above it and both record trays in the bay below. The record ledge **wraps round onto the side wall**, with the PC on a shelf above the end of the L-sofa. The chimney breast gets a **fold-down oak dining table** for four (six or eight with it lifted off), and a **second spotlight bar** goes on your existing ceiling point by the door. The **sub** moves under the dining wall, on the front wall beside the TV.

1 · The plan on a page



Drawn to your measurements, in Card Room Green: the alcove with the turntable zone on the right, the dining wall folded on the chimney breast, both spot bars. Full-resolution renders come with this plan.

£2,532–4,772
plus the electrician, item by item in section 11

41 of 53
fit checks pass outright; 8 are tight, 4 need a check on site (section 3)

188 · 240 · 4–8
LPs and singles, all face-out · seats at the table

Area	Decision
Alcove	The TV between two oak uprights, the left one now a corner post on the side wall; on the right, the turntable on the worktop and the Series X on a shelf at 134; below, three 60 cm bays: the Xbox consoles, the PlayStations and the two record trays.
Side wall	The record ledge wraps the corner with four more LPs, then an oak shelf for the PC above the end of the sofa, clear of the TV's sightlines; the sub on the floor by the wall.
Dining wall	A 110 × 80 solid oak table on an oak backboard: folded flat every day, down on two Häfele Hebgo brackets for four, lifted off onto legs for six, with a leaf for eight.
Records	Every record face-out: about 188 LPs (180 in the trays, 8 on the two ledges) and 240 singles.
Heights	Worktop and table 76 · TV centre 116.5 · Series X shelf 134 · ledges and PC shelf 155.4 · high shelf 196, 39 under the 235 ceiling.
Colour and light	Card Room Green No. 79, colour-drenched. The ceiling LED strip, LED grooves in the uprights, a bias light, and two Wolston-style spot bars on your two existing ceiling points: between the TV and the sofa, and between the door and the wall.
Power and network	The side-wall socket becomes a double: left for the 10-gang (everything modern), right for the retro set, switched at the wall. A new socket behind the sofa for the PC and the rears. Deco A in bedroom 3, Cat6 down the airing cupboard.

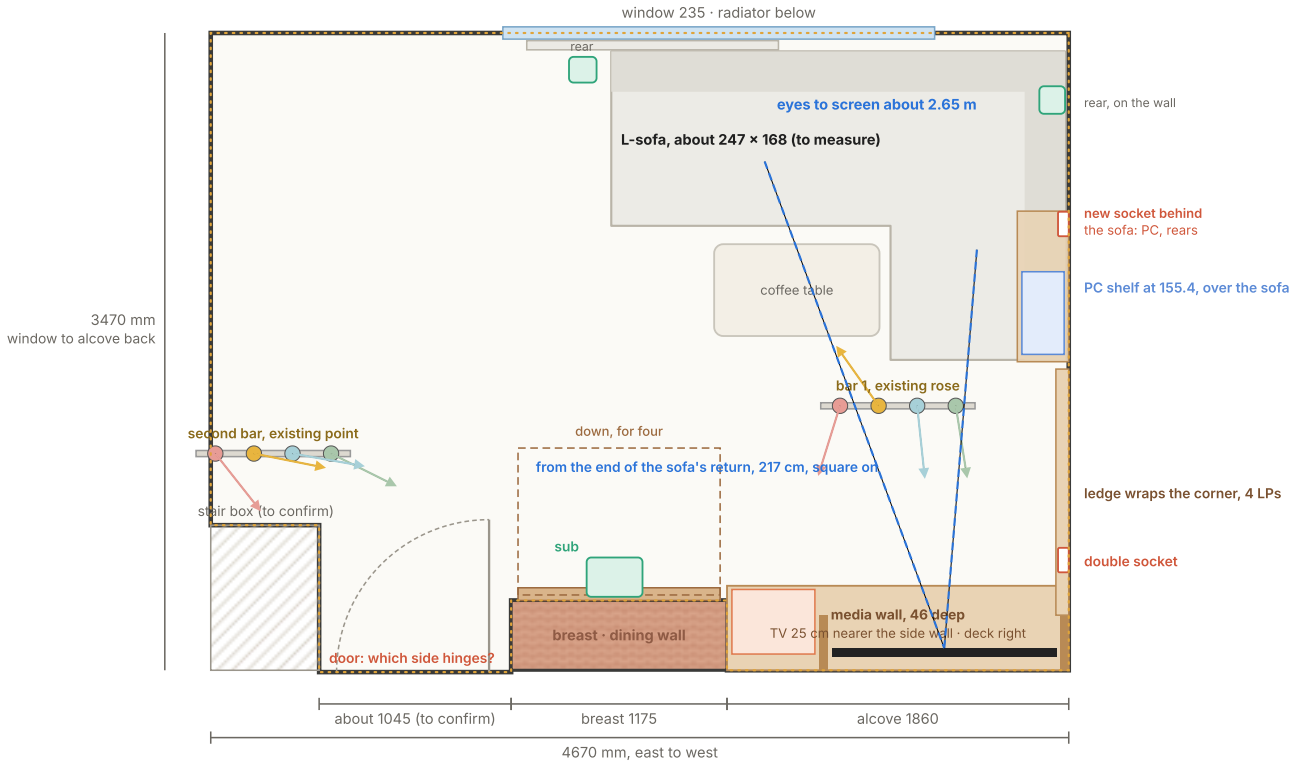
To confirm (full list in section 12): the L-sofa's size · where the door to the hall hinges, and the west end's sizes · the turntable, PC case and fan stand models · chairs you already have · where the leaf and legs live · the 23 × 22 cm box by the window.

2 · The room

Everything is drawn from your hand-measured sketch; where the agent's floor plan disagrees, your tape wins. The L-sofa is drawn at a typical 247 × 168 cm (a corner sofa of that size[23]) until you measure yours.

The living room from above, as you measured it

To scale, north at the top. Media wall in the alcove, the ledge and PC shelf on the side wall over the sofa, the dining wall on the chimney breast, two spot bars



Ceiling 2350 mm. The L-sofa is drawn at a typical 247 × 168 cm until you measure yours; the west end (stair box, door) is drawn roughly until you confirm it.

From your sketch	mm	Notes
Room, east to west along the window	4670	The media wall is the south wall, shared with the hall and garage
Window wall to the alcove's back wall	3470	To the chimney breast's face: 3090
Ceiling	2350	
TV alcove, east wall to the breast	1860 × 380	Floor to ceiling
Chimney breast	1175	Projects 380 on the alcove side, 390 on the west side: the dining wall
Window and radiator (north wall)	2350 · 1370	Window starts 730 from the east wall; radiator ends 130 inside its west end
Skirting	60 × 20	White now; Card Room Green eggshell in the plan
West end: recess, door, stair box	935 / 1045 / 660 / 820	Drawn roughly: please confirm which is which

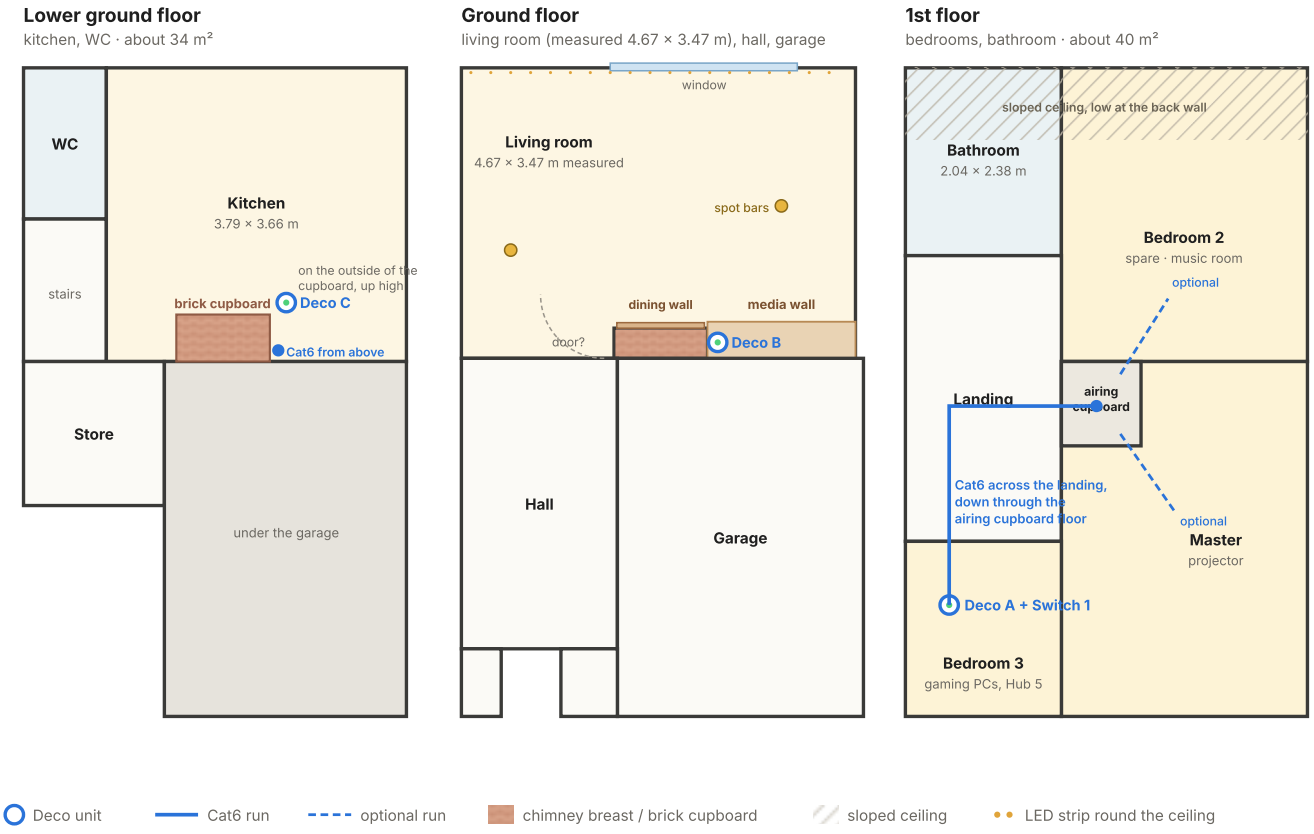
From the sofa: from the middle of the long side your eyes are about 2.65 m from the screen, which a 55-inch fills at about 26°: right for TV and games. With the TV 25.25 cm nearer the side wall, it now sits square

on to the end of the sofa's return (217 cm away), and only about 12° off-centre from the middle of the long side. The screen's centre at 116.5 cm is 1–2.5° above a seated eye line. Both spot bars hang 22.4 cm, so the lowest shade is at 212.6 cm: no headroom worries.

The house

The house: where the Decos and Cat6 runs go

Deco A and Switch 1 move to bedroom 3 with the Hub 5. Both Cat6 runs drop through the airing cupboard, right over the media wall's corner



Outline traced from the agent's plan (north, the window side, at the top). The living room's alcove, chimney breast and windows are placed from your measurements.

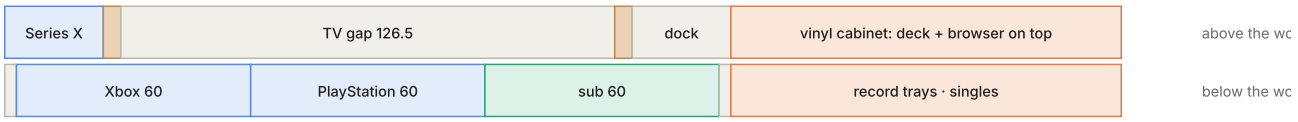
Deco A and Switch 1 sit with the Hub 5 and the gaming PCs in bedroom 3; both Cat6 runs go across the landing and drop through the airing cupboard, which sits over the alcove's corner (section 9).

3 · The media wall

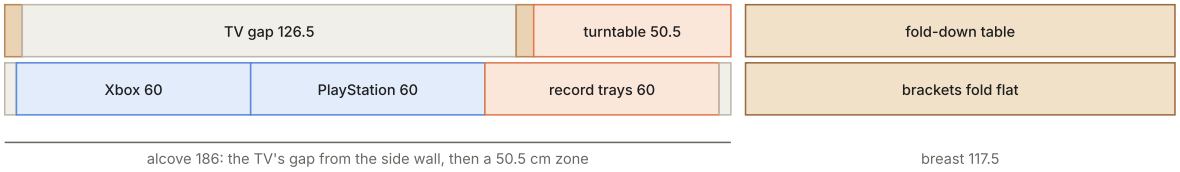
What moved in v11

To scale, from the side wall. Top: v10. Bottom: v11

v10: the TV in the middle, the vinyl cabinet on the breast



v11: the TV 25.25 cm nearer the side wall, the dining wall on the breast



- Moving the TV 25.25 cm left joins the two 25.25 cm side gaps into one 50.5 cm zone: wide enough for a 45.2 cm turntable, with 2.65 cm each side.
- The Series X moves up onto a shelf over the deck, above the lid's swing, with 15 cm of air each side. The records sit in bay 3, right under the deck.
- That frees the chimney breast for a fold-down dining table; the sub, which is wireless, moves to the floor by the side wall.

Why this layout, and what we ruled out

- **The TV in the middle (v10):** left two 25.25 cm gaps either side, too narrow for a turntable, so the vinyl had to move out to the chimney breast. Moving the TV joins them into one 50.5 cm zone.
- **The turntable with the Series X beside it:** 45.2 + 20.5 cm won't go in 50.5. The Series X goes up on a shelf instead, above the lid's swing, where it gets more air than it had in v10 (15 cm each side, 21 above).
- **The vinyl cabinet on the breast as well as a table:** a table needs the breast's full width, and a cabinet in front would put the diners 44 cm further into the room.
- **A gateleg table in the recess:** it would block the door; the breast is the only clear wall.



Numbered view: the game drawer and the lower record tray are pulled out 30 cm, with a view of each from above. Dashed lines show what's hidden: gold for data, audio and low-voltage leads, coral for mains. The sofa isn't drawn so the base stays in view; the room plan shows it.

OAK JOINERY	
1	Solid oak uprights, 4.5 × 30, worktop to ceiling (159 long)
2	The left upright is a corner post against the side wall
3	Flush wedged tenons carry the Series X shelf (134)
4	High shelf, top 196: books up to 30 cm
5	Record ledge above the TV: 4 LPs face-out
6	The ledge wraps the corner: 4 more LPs face-out
7	LED groove in each upright washes the wall
8	High shelf, right: the “now playing” 7-inch
ON THE WORKTOP	
9	LG OLED C6 55in · centre 116.5 · now 25 cm nearer the side wall
10	Soundbar · eARC, optical and AUX inputs
11	Turntable (AT-LP120X size allowed) · lid clear to 130
12	Xbox Series X on its fan stand · 15 cm air each side
13	Phono preamp, hung under the shelf at the back
14	Solid oak worktop 186 × 46 × 3, top at 76
IN THE ALCOVE BASE	
15	Oak face frame; its bottom rail hides the 9 cm plinth
16	Xbox One + power brick · bay 1 top row
17	Xbox 360 + power brick · bay 1 middle row
18	Game drawer: 2 face-out bins, spine-up rows behind
19	PS3 + controller basket · bay 2 top row
20	PS2 + HDMI switch · bay 2 middle row
21	PS1 + RetroTINK · bay 2 bottom row
22	Upper record tray: ~90 LPs + ~120 singles · bay 3
23	Lower record tray: the same, shown pulled out
24	Hidden slim drawer in the plinth, push to open

THE DINING WALL ON THE CHIMNEY BREAST	
25	Oak backboard: lift-off hinges on top, brackets below
26	Oak table 110 × 80, pegged breadboard ends: folds up flat
27	Häfele Hebgo folding brackets, 500 kg a pair
28	Turn buttons hold the folded table
29	Two framed LPs above
30	Subwoofer on an oak riser, between the brackets
THE SIDE WALL	
31	PC on its oak shelf, above the end of the sofa (see the room plan)
32	Oak frame with woven cane hides the PC's ports
33	Pad charging dock
34	PC cables: trunking down to the skirting
POWER, NETWORK AND LIGHT	
35	Deco B: leads in the groove behind the right upright
36	Double socket: its right switch is the retro switch
37	10-gang extension, hidden behind the TV
38	LED strip round the ceiling edge (as now)
39	Spot bar on the existing rose, between TV and sofa
40	Second spot bar, existing point by the door (just out of view)

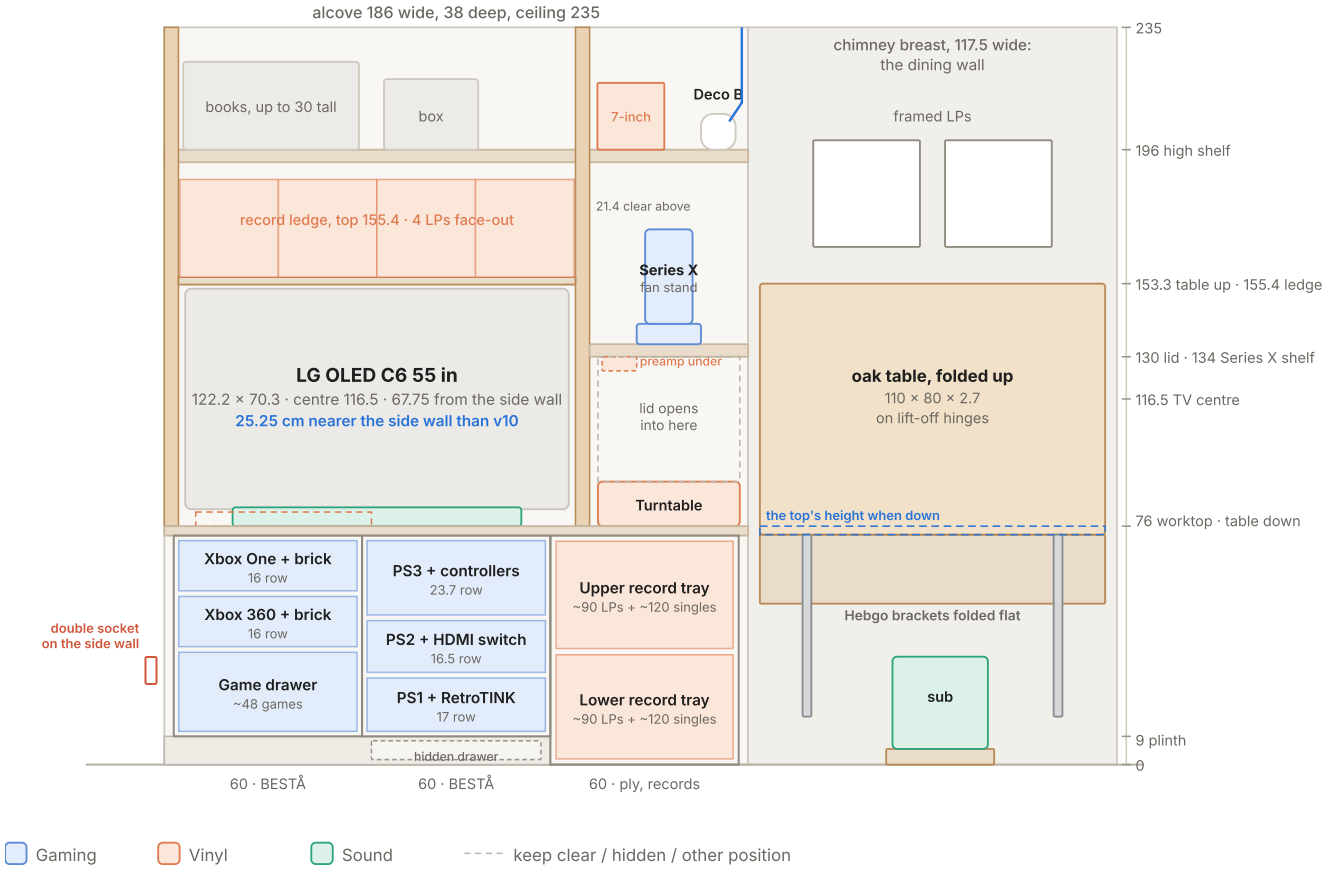
Built to your alcove: 186 wide, 38 deep, 235 to the ceiling

- **Base:** 3 cm filler, three 60 cm bays, 3 cm filler. Bays 1 and 2 are IKEA BESTÅ 60×40×64 frames on a 9 cm plinth (tops at 73, back panels off); bay 3 is an 18 mm ply carcass 60 × 40 × 73 from the floor, for the record trays. One oak face frame runs across all three; its bottom rail hides the plinth and a push-to-open hidden drawer under bay 2.
- **Depth:** 40 cm frames, a 4 cm cable channel behind and the 2 cm face frame make 46 cm from the back wall, 8 cm proud of the chimney breast.
- **Worktop:** solid oak, 186 × 46 × 3, top at 76, with a brush-lined slot behind the TV and a grommet behind the deck.

- **Oak uprights:** 4.5 × 30 cm, 159 cm from the worktop to the ceiling. The left one stands against the side wall at 0–4.5, a corner post that frames the TV and carries the wrap ledge. The right one is at 131–135.5. That leaves 126.5 cm between them, 2.15 cm clear of the TV each side.
- **TV:** centre 67.75 cm from the side wall (it was 93), on LG's stand set 2 cm off the back wall: bottom 81.4, centre 116.5, top 151.7. The soundbar sits centred under it at the worktop's front edge.
- **The turntable zone:** 50.5 cm between the right upright and the breast. The deck sits on the worktop with 2.65 cm each side and its lid opens to about 121. Above it is an oak shelf at 134, through-tenoned into the upright, carrying the Series X on its fan stand; the phono preamp hangs underneath at the back.
- **Shelves above:** the record ledge at 155.4 between the uprights, four LPs face-out; the high shelf at 196, wall to wall in two lengths, with books, a box, a print, the framed "now playing" 7-inch and Deco B.
- **Power:** the side-wall socket in front of bay 1 becomes a double. Its two leads enter through a notch at the left end of the face frame's bottom rail and run back through the plinth to the channel.

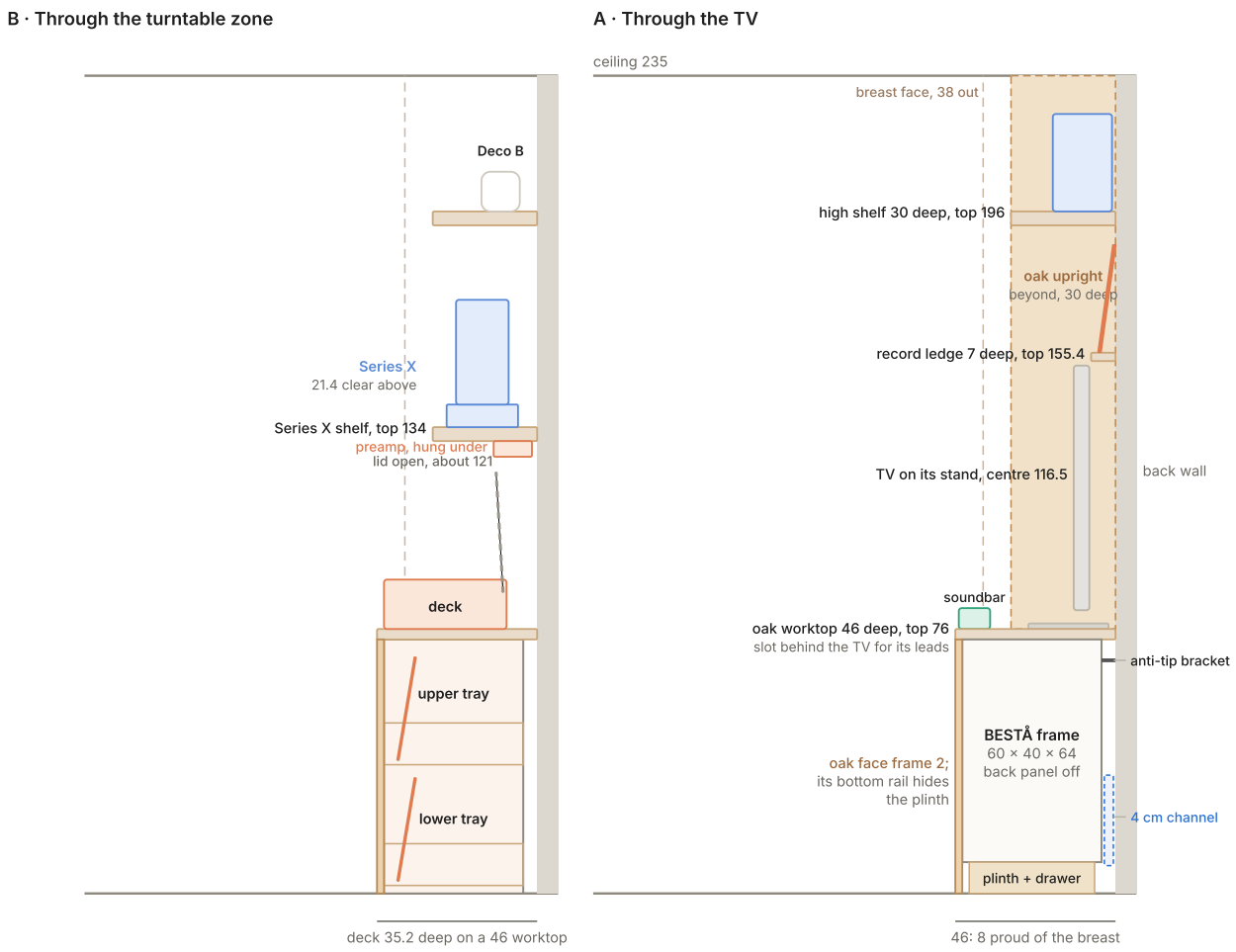
The media wall and the dining wall from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below; the fold-down table on the chimney breast. Heights in cm



How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



The fit proof: every item against its published size

Each zone below lists what goes in it, its size and where that size comes from (numbered sources, section 12), the space, what's left and a verdict. 41 of the 53 checks fit outright and none fail. 8 are tight: the deck's 2.65 cm sides, the preamp over the open lid, the dining seats at 55–57 cm each, and the clearances for six and eight. 4 need a check on site: the TV's plugs by the corner post, the RetroTINK's size, the PC case, and the door's swing.

The TV between the uprights

126.5 between the oak uprights, the left one now a corner post on the side wall; the ledge's underside at 153

Item	Size (cm)	Space	Spare (cm)	Result
LG OLED C6 55 in, on its stand	122.2 × 70.3 × 4.5 panel; 75.7 tall on the stand [2]	126.5 wide	2.15 each side	**Fits**
Top of the screen	151.7 (bottom 81.4, centre 116.5) [2]	ledge at 153	1.3	**Fits**
LG stand base	47 × 23 [2]	worktop 46 deep	set 2 cm off the wall: 11 cm in front of it for the soundbar's leads	**Fits**
Soundbar, Hisense AX5100G, centred under the TV	92 × 9 × 6 tall [3]	x 21.75–113.75, top 82	the sightline to the screen's bottom passes 83.54–84.17 over it	**Fits**
The TV's HDMI plugs, with the screen 2.15 from the corner post	C6 sockets face sideways on the back panel [2]	behind the screen	plugs stay behind the panel; if a stiff lead pushes out, use 90° HDMI adapters	**Check**

The turntable zone, right of the TV

50.5 wide × 46 deep, worktop 76 to the Series X shelf's underside at 130

Item	Size (cm)	Space	Spare (cm)	Result
Turntable, AT-LP120X size (worst case)	45.2 × 35.2 × 14.2 with its lid [8]	50.5 wide	2.65 each side	**Fits, tight**
Its depth, with room behind for the lid's hinges	35.2, front 2 cm in from the worktop's edge [8]	46 deep	8.8 behind it	**Fits**
Lid open	cover 44.1 × 34.6; standing up, its top at about 121 [17]	shelf underside 130	about 9	**Fits**
Phono preamp, hung under the shelf at the back	about 11 × 11 × 4.5	125.5–130, at the back	4.4 over the open lid's edge; lift the lid gently	**Fits, tight**
Series X fan stand on the shelf	20.5 × 20.5 (assumed; model to confirm)	50.5 × 30 shelf	15 each side	**Fits**
Xbox Series X, standing	15.1 × 15.1 × 30.1 tall [4]	top at 170.6	21.4 clear above (Microsoft's minimum is 4); 15 each side, inside the 10–15 it recommends	**Fits**
Deco B on the high shelf, right	11 × 11 × 11.4 [7]	50.5 wide, 39 to the ceiling	beside the framed 7-inch, 27 clear above	**Fits**

Shelves above the TV

ledge top 155.4, 7 deep; high shelf top 196, 30 deep; ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Four LPs on the record ledge	4 × 31.4 = 125.6 wide[13]	126.5 long	0.9: snug, as drawn	**Fits**
Ledge LPs leaning back 8°	tops at 186.5[13]	high shelf underside 192	5.5 to lift one out over the 1.5 lip	**Fits**
Books on the high shelf	up to 30 tall	39 to the ceiling	9 over the tallest	**Fits**
Oak uprights	159 long (worktop to ceiling)[1]	235 ceiling	buy 165 and scribe to the ceiling	**Fits**

Bay 1, Xbox (BESTÅ frame on the 9 cm plinth)

56.8 wide inside, 40 deep, rows as in v9

Item	Size (cm)	Space	Spare (cm)	Result
Xbox One, original, and its brick	33.3 × 27.4 × 7.9; brick 17 × 7.5 × 4.8[12][11]	16 tall row	8.1 above; 21.5 beside the console for the brick	**Fits**
Xbox 360, original, and its brick	30.9 × 25.8 × 8.3; brick about 21 × 10 × 6[12][16]	16 tall row	7.7 above; 25.9 beside for the brick	**Fits**
Game drawer	box 54.3 × 35 on 350 mm runners; cases 19 × 13.5 × 1.4[15][13]	25.2 tall row, 40 deep	about 3 above the leaning cases; 5 behind the box	**Fits**
Depth with the leads	console 27.4 + HDMI plug about 5[12]	40 deep, back open	about 7.6, then the 4 cm channel	**Fits**

Bay 2, PlayStation (BESTÅ frame on the plinth)

56.8 wide inside, 40 deep

Item	Size (cm)	Space	Spare (cm)	Result
PS3, original, + controller basket	32.5 × 27.4 × 9.8[12]	23.7 tall row	13.9 above; 24.3 beside for the basket	**Fits**
PS2, original, + HDMI switch	30.2 × 18.3 × 7.8; switch 12.7 × 5.9 × 2.5[12][10]	16.5 tall row	8.7 above; 26.6 beside	**Fits**
PS1, original, + RetroTINK-6X CE	27 × 18.8 × 6; RetroTINK size not published[12][9]	17 tall row	29.8 beside the PS1 reserved for it; measure it when it arrives	**Check**
Hidden slim drawer in the plinth	box 52 × 34 × 5, push-to-open 300 mm runners[15]	9 cm plinth, 120 wide	about 2 above and below the box	**Fits**
Switch 2 (TL-SG108) and the retro 6-way in the channel	15.8 × 10.1 × 2.5; 6-way about 34 × 6 × 4[6]	4 cm channel	on their edges	**Fits, tight**

Bay 3, the records (18 mm ply carcass from the floor)

56.4 wide × 40 deep × 69.4 tall inside, under the turntable

Item	Size (cm)	Space	Spare (cm)	Result
Two face-out record trays	trays 53.9 × 40 on 400 mm runners, 68–70 kg pairs [15] [13]	56.4 inside	exact: 53.9 + 2 × 1.25 runner allowance = 56.4	**Fits**
Tray heights	lower sleeves to 34.4, upper to 69.1 [13]	top panel underside 71.2	2.6 and 2.1 finger room	**Fits**
LP trough · singles trough	32.6 for 31.4 sleeves · 18.9 for 18.3–18.4 [13]	52.1 inside each tray	0.6 each side · 0.25–0.3 each side	**Fits**

The side wall: the ledge wraps the corner, the PC shelf

one oak line at 155.4: ledge 134 long × 7, then a shelf 82 × 28

Item	Size (cm)	Space	Spare (cm)	Result
Four more LPs on the wrap ledge	4 × 31.4 + gaps = 129.8 [13]	134 long	2.7	**Fits**
The PC on its shelf	23 × 45 × 47 tall (a typical mid-tower; model to confirm)	28 × 82	2.5 off the wall, the cane screen and pad dock beyond it	**Check**
Air above the PC	top at 202.4	235 ceiling	32.6 for the top exhaust	**Fits**
Shelf over the end of the sofa	underside 151.4 [23]	sofa back 80	71.4; the shelf covers only the 28 cm nearest the wall	**Fits**
Out of the TV's sightlines	PC about 54° left of the line to the screen from the sofa [1]	—	v10's PC sat at the screen's left edge	**Fits**

The dining wall on the chimney breast

breast 117.5 wide; the table 110 × 80 × 2.7 solid oak, about 17 kg

Item	Size (cm)	Space	Spare (cm)	Result
Table on the breast	110 wide [1]	117.5	3.75 each side	**Fits**
Folded up	about 6.9 off the breast, top edge 153.3 [19]	framed LPs from 165	11.7	**Fits**
Two Hebgo brackets under it	arms 58, 80 apart; 500 kg a pair [18]	a 80 deep top	22 beyond the arms; spacing inside Häfele's 80	**Fits**
Three lift-off hinges on the backboard	100 × 88, rated for a 120 kg door [19]	a 17 kg top	slide it 5 cm sideways to lift it off	**Fits**
Four at the wall: two on the room side	110 of edge [22]	60 each recommended	55 each	**Fits, tight**
Four at the wall: room to the sofa	table edge 83 out [22] [23]	sofa front	121 (walk behind: 100+)	**Fits**
The west end chair and the door	chair 43 × 49 [21]	the door's swing, if it opens in	they overlap: keep the door shut while you eat, or confirm it opens outwards	**Check**
Six: lifted off onto four legs	top at 73.7–75.7 [20]	dining height about 75	within 1.3	**Fits**
Six: two a side	110 long [22]	60 each	55 each	**Fits, tight**
Six and eight: to the breast and the window	table 60–140 out [22]	60 minimum	60 to the breast (tight minimum); 169 to the window	**Fits, tight**
Six: the west end	table end at 365 [22]	west wall at 467	102	**Fits**
Eight: with the 60 × 80 leaf	170 long, three a side [22] [20]	60 each	56.67 each	**Fits, tight**
Eight: the west end and the sofa	table 220–390 [22] [23]	75 to aim for	77 to the wall; 64 to the sofa's front at the east end	**Fits, tight**
Sub under the dining wall, on its side	30.4 long × 21.4 out × 29.4 tall, on a 5 cm riser [3]	between the bracket arms, 80 apart	25.85 to the east arm; 20.95 behind the rear port	**Fits**
Feet under the table, four at the wall	sub 25.4 out from the breast [21]	table 80 deep	57.6 of legroom in front of it on the room side; the end seats are clear	**Fits**
Chairs at the table	seat 48 [21]	top 76	28 seat to top, the usual 25–30	**Fits**

The room

467 × 347, ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Eyes to screen from the sofa	the L-sofa's long side against the window wall, 10 off the radiator[1][2][3]	335 window wall to screen	about 2.65 m; a 55-inch fills about 26°	**Fits**
From the end of the sofa's return	sitting up at its end, 217 from the screen[2][3]	straight ahead of the TV	the TV's move puts it square on to this seat	**Fits**
TV centre against seated eyes	116.5[2]	eyes at 105–112	about 1–2.5° above the eye line	**Fits**
Both spotlight bars under the ceiling	22.4 drop[1][4]	235 ceiling	lowest shade at 212.6	**Fits**

Console sizes, every version

The bays are sized for the largest version of each console. The model number is on the label underneath; a smaller version just leaves more room.

Console	Version	W × D × H (cm)	Power
PS1	Original (largest) · PS one	27.0 × 18.8 × 6.0 · 19.3 × 14.4 × 3.8	Built in · external adapter
PS2	Original (largest) · Slim	30.2 × 18.3 × 7.8 · 23.1 × 15.2 × 2.8	Built in · external brick
PS3	Original (largest) · Slim · Super Slim	32.5 × 27.4 × 9.8 · 29.0 × 29.0 × 6.5 · 29.0 × 23.0 × 6.0	Built in
Xbox 360	Original (largest) · S · E	30.9 × 25.8 × 8.3 · about 27.0 × 26.4 × 7.5 · 25.4 × 26.4 × 6.7	External brick
Xbox One	Original (largest) · S · X	33.3 × 27.4 × 7.9 · 29.5 × 22.9 × 6.4 · 30.0 × 24.0 × 6.0	External brick · built in
Xbox Series X	Standing	15.1 × 15.1 footprint, 30.1 tall	Built in

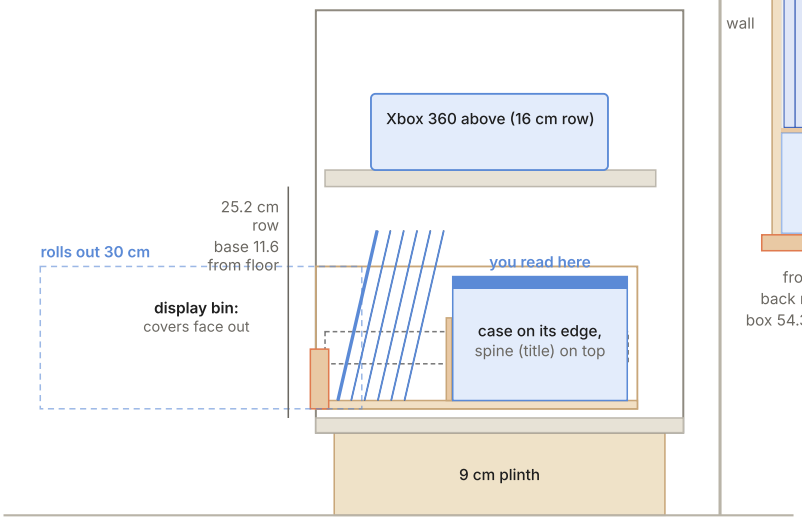
The game drawer (bay 1, bottom row)

Unchanged. Two display bins at the front show covers face-out even with the drawer shut (Xbox left, PS2 right, a small tray between them); behind them the rest of the games stand on edge with the spine on top, so every title faces up when you roll it out.

The game drawer: covers on show at the front, every title readable from above at the back

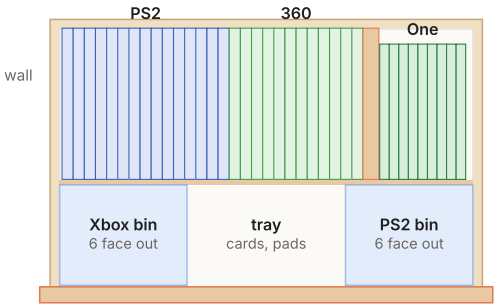
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out



front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

- **Box:** about 54.3 × 35 cm outside on 350 mm full-extension soft-close runners screwed to the BESTÅ sides; sides 15 cm; 9 mm base lined in felt. About 48 games.
- **Runners:** Accuride DZ3832EC (43 kg) or Screwfix Essentials 964VN (40 kg); not IKEA's BESTÅ runners, which aren't full extension.
- **The sub:** it sits under the dining wall now, so nothing stands in front of the game drawer.

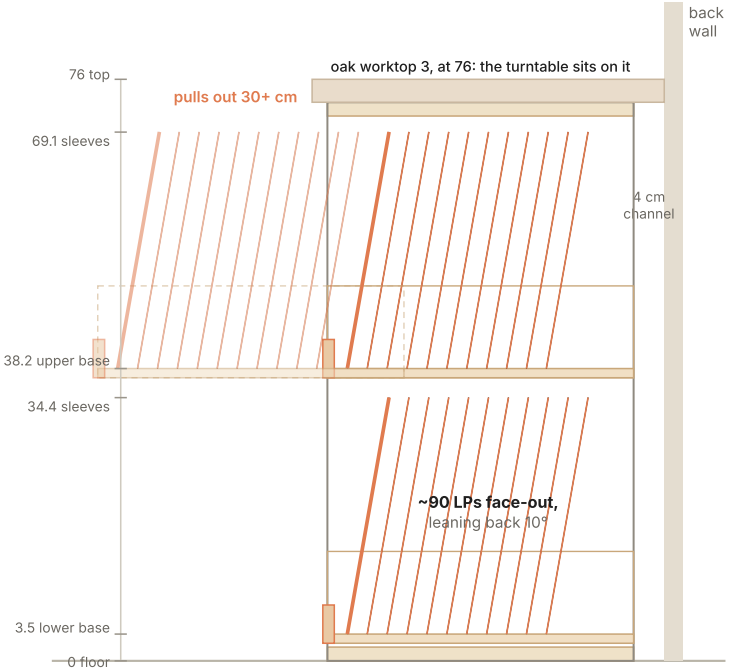
4 · The records and the turntable

The vinyl is back in one place, in the alcove: the deck on the worktop, the records in the two trays right underneath it, the preamp tucked under the shelf above. You can pull a tray, pick a record and play it without crossing the room.

The record trays: every LP and single face-out, two tiers in bay 3

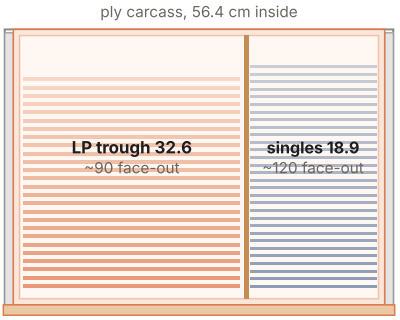
Bay 3 of the alcove base, right under the turntable. Section left, plan right. Pull a tray and flip through the covers like a record shop browser

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · trays 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Top view of one tray



tray 53.9 × 40 outside, 52.1 × 38.2 inside
9 mm sides · 6 mm oak divider · 5 cm oak lip
about 32 kg full: open one tray at a time

How the record trays work

- **LP trough:** 32.6 cm wide, so a bare 31.4 cm sleeve has 6 mm each side. Records lean back 10° like a record shop browser; the last one leans on a raked oak rest.
- **Singles trough:** 18.9 cm wide, the same lean.
- **Capacity:** about 90 LPs and 120 singles per tray; two trays, about 180 LPs and 240 singles.
- **Runners:** 400 mm full extension, 68–70 kg per pair, screwed straight into bay 3's 18 mm ply sides. A full tray weighs about 32 kg: open one at a time, and fit the anti-tip brackets before loading.
- **Outer sleeves:** if most LPs live in thick outer sleeves (up to 32.3 cm), widen the LP trough to 33.4 cm; the singles trough then drops to 18.1, too narrow for face-out singles.

Dimension	Figure	How it adds up
Bay 3 carcass	60 × 40 × 73 outside, 56.4 × 69.4 inside	18 mm birch ply; back open to the channel
Tray	53.9 × 40 outside, 52.1 × 38.2 inside	56.4 − 2 × 1.25 runners; 9 mm sides and ends
Inside layout	32.6 + 0.6 + 18.9	LP trough, oak divider, singles trough
Lower tray · upper tray	Undersides 2.3 · 37.0; sleeves to 34.4 · 69.1	2.6 and 2.1 cm finger room; top panel at 71.2

The turntable zone

- **The deck:** on isolation feet, 2 cm back from the worktop's front edge, with 8.8 cm behind it for the lid's hinges. The zone takes a deck up to the size of an AT-LP120X (45.2 × 35.2 cm); most decks are smaller.
- **The lid:** AT's cover is 44.1 × 34.6 cm[17], so standing open its top edge is at about 121; the shelf's underside is at 130.
- **Bass:** the sub stands under the dining wall on its own oak riser, off the base the deck sits on; isolation feet under the deck take care of the rest.
- **Signal path:** turntable set to PHONO → the preamp under the shelf → a 2 m RCA-to-3.5 mm lead down the groove behind the right upright and along the back of the worktop to the soundbar's AUX. If the deck has no PHONO/LINE switch, plug it straight into the AUX.
- **Soundbar:** Music mode, bass up, surround, night and dialogue modes off. Always the AUX cable, never Bluetooth.

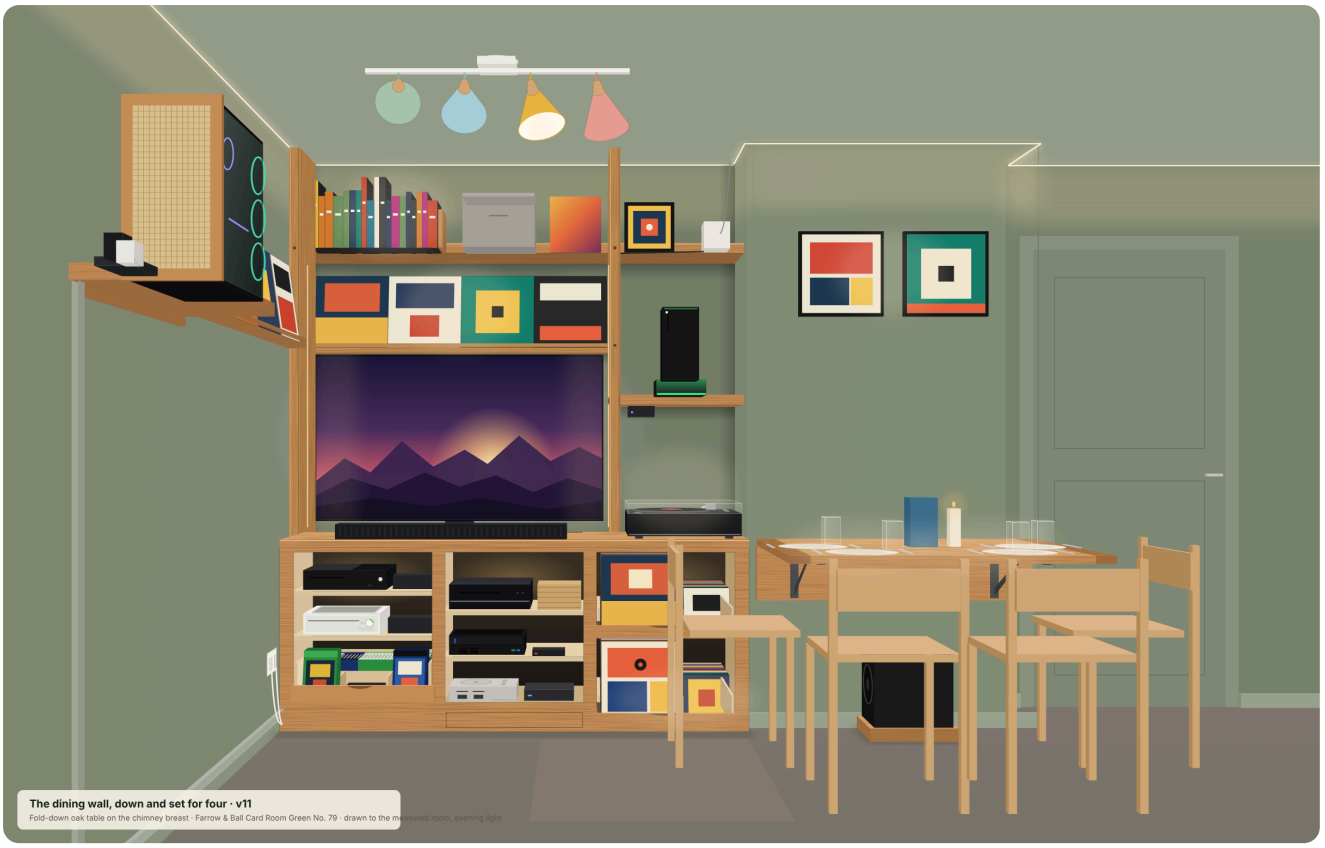
Where every record lives

Where	What	How many
Bay 3, upper and lower trays	LPs and singles, face-out	About 180 LPs + 240 singles
Above the TV	Oak record ledge, the "now playing" wall	4 LPs
Side wall	The ledge wrapped round the corner	4 LPs
High shelf, right	A framed 7-inch	1
Total		**About 188 LPs and 240 singles, every one face-out**

That's fewer than v10's cabinet held (it had a 50-LP browser and a singles drawer). If the collection is bigger than this, tell me the counts. The neatest overflow is a matching face-out tray unit under the window at the end of the sofa.

5 · The dining wall

The chimney breast becomes the dining wall. A solid oak table, 110 × 80 cm, hangs on lift-off hinges on an oak backboard. Every day it stands folded flat against the breast, about 6.9 cm deep. Down, it's a table for four at the same 76 cm as the alcove's worktop. Lifted off onto four legs, it seats six in the room, or eight with a matching leaf.

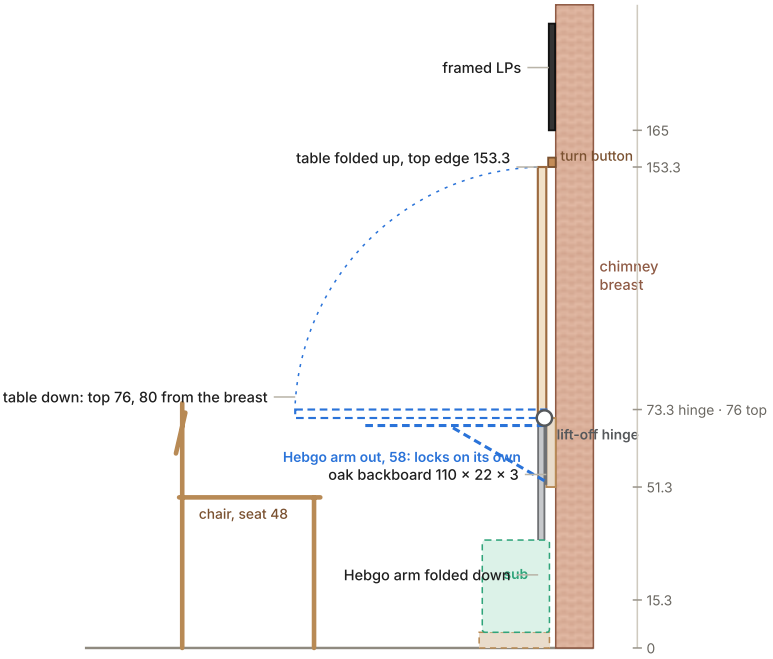


Down and set for four: two on the room side, one at each end, the sub tucked under the wall. The second spot bar, on the point by the door just out of view, lights the table and the framed LPs.

The dining wall in section: folded, down, or off the wall

Through the chimney breast at the table's centre, to scale, room on the left. Solid: folded. Dashed blue: down for four

- Everyday: folded flat, about 7 cm off the breast, held at the top by two turn buttons.
- For four: lift the top edge a little, turn the buttons, lower it onto the two brackets as they swing out and lock.
- For six or eight: with it down, slide it 5 cm sideways off the lift-off hinges' pins, screw in four legs (710–730 mm),
- and stand it in the room: the top is then 73.7–75.7 cm. Two people lift it: about 18 kg.



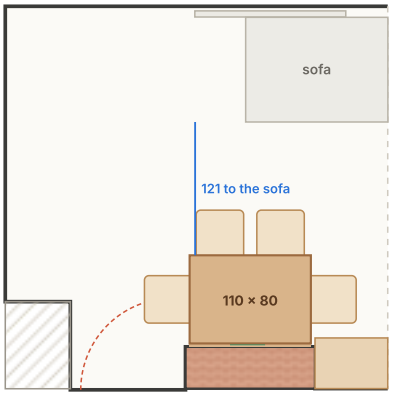
How it works

- **The top:** solid oak with **pegged breadboard ends**, 7 cm oak strips across each end on a tongue, pinned with three oak pegs in slotted holes so the top can move. They stop it cupping when it lives upright, and they frame it like the uprights frame the TV. The corners are rounded to a 5 cm radius (kinder on hips by a door), and the underside edges are chamfered 1 cm so it reads slim.
- **Backboard:** oak, 110 × 22 × 3, from the worktop's offcut, screwed to the breast with plugs every 20 cm, top at 73.3.
- **Brackets:** two Häfele Hebgo heavy-duty folding brackets (287.43.428: 580 mm arm, 500 kg a pair[18]), 80 cm apart on the backboard's face. Raised, they lock on their own; a light lift on the arm releases them, and they fold flat down the breast.
- **Hinges:** three 100 mm lift-off hinges[19], all the same hand, along the backboard's top edge. Slide the table 5 cm sideways and it comes off.
- **Folded:** two oak turn buttons on the breast, plus a magnetic catch, hold the top edge at 153.3. The underside faces the room, so it's oiled like the top; the four leg plates show as small dark discs.
- **For six or eight:** lift it off (two people, about 18 kg), screw four legs into the plates: tapered oak legs (4.5 cm to 3 cm, on M8 hanger bolts) to match the wall, or Emuca 710 mm steel legs[20] (adjustable to 730) as the cheaper option, and stand it in the room. For eight, butt the 60 × 80 leaf, on its own four legs, against one end, joined by two alignment dowels and two table clips.
- **Chairs:** drawn at IKEA TERJE size (43 × 49, seat 48[21]), which folds flat. Any dining chair with a 45–48 cm seat works.

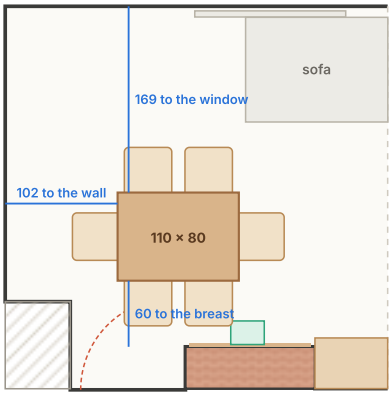
The dining table for four, six or eight

To scale, the west half of the room from above, window at the top. Red: the door's swing. Blue: the clearances that matter

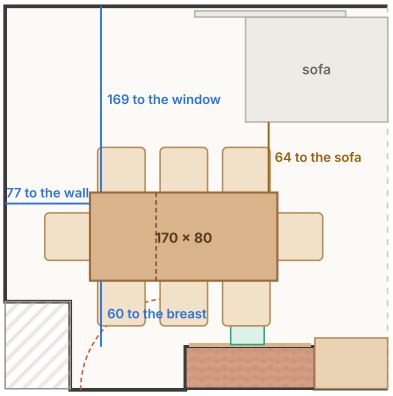
Four, at the wall
everyday: the table down on its brackets



Six, in the room
lifted off its hinges onto four legs



Eight, with the leaf
plus a 60 × 80 leaf on four more legs



- Per diner: 60 cm of table edge (55 at a squeeze). Table edge to a wall: 60 cm is a tight minimum, 75 to aim for, 100+ to walk behind someone seated.
- Six and eight: slide the coffee table to the window first. The dashed red arc is the door's swing if it opens into the room: the chairs on that side need it shut while you eat.
- Chairs drawn at IKEA TERJE size (43 × 49, seat 48): 28 cm from seat to table top, the usual 25–30.

Seats	Where the table stands	Per diner	Clearances
Four	Down at the wall, on its brackets	55 on the room side (60 is ideal); the ends are 80 wide	121 to the sofa: room to walk behind
Six	Lifted off, centred at about 310 from the side wall, 60 off the breast	55, two a side	60 to the breast (the tight minimum), 169 to the window, 102 to the west wall
Eight	Plus the leaf: 170 × 80	57, three a side	77 to the west wall, 64 to the sofa at the east end: tight but fine for a dinner

Clearances from a UK planning guide: 60 cm of table edge per diner, 60 cm from the table to a wall as a tight minimum, 75 to aim for, 100–110 to walk behind someone seated[22]. For six or eight, slide the coffee table to the window first, and keep the door to the hall shut while you eat if it opens into the room: the chairs on that side sit in its swing.

Cut list and hardware

Part	Material	Size (cm)	Qty
Table top · leaf	Solid oak, edge-glued staves	96 × 80 × 2.7 centre + two 7 × 80 breadboard ends (110 overall) · 60 × 80 × 2.7	1 · 1
Breadboard pegs	10 mm oak dowel, slotted holes in the tongue	3 per end	6
Legs (oak option)	Solid oak, tapered on two inside faces	71 × 4.5 tapering to 3, M8 hanger bolt and angled plate	8
Backboard	Solid oak, from the worktop offcut	110 × 22 × 3	1
Turn buttons	Oak offcut	6 × 2.5 × 2.2, on a single screw	2
Sub riser	Oak offcut, felt feet	34 × 25 × 5	1

Folding brackets	Häfele Hebgo 287.43.428	arm 58, height 22	2
Lift-off hinges	Union PowerLoad, 100 × 88	one 3-pack, same hand	3
Legs (steel option)	Emuca Ø60 steel, 710–730 mm, with plates	—	8
Leaf joint	Alignment dowels, table clips	—	2 + 2

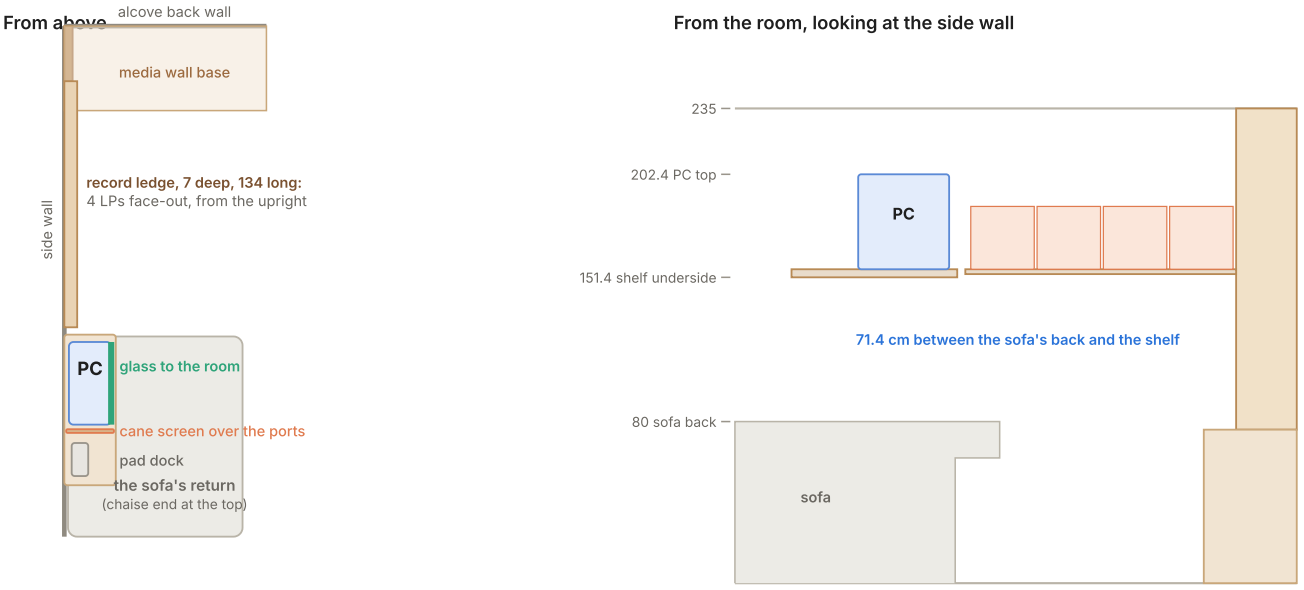
Safety: never fold it up with anything on it, keep both turn buttons turned when it's up, and check the brackets have locked (a click each side) before you load it.

6 · The side wall

One oak line at 155.4 cm runs from the ledge over the TV, round the corner on the left upright, and along the side wall: first a 7 cm record ledge with four more LPs face-out, then an oak shelf for the PC over the end of the sofa. It ties the alcove to the rest of the room and moves the PC out of the TV's sightlines.

The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines



- **Wrap ledge:** 134 × 7 × 2.4 cm with a 1.5 cm lip, on an oak cleat screwed to the side wall, butt-jointed to the left upright's front edge on two 10 mm dowels.
- **PC shelf:** 82 × 28 × 4 cm, on an oak cleat with two concealed heavy-duty brackets (40 kg+ a pair), 71.4 cm above the sofa's back. The case sits with its glass to the room and its front towards the alcove. The ports face the window behind an oak frame with a woven cane panel, open at the top for the exhaust. The pad dock sits beyond the screen.
- **Cables:** through a notch at the shelf's back corner, down slim trunking painted Card Room Green, along the skirting behind the sofa and into the plinth. The HDMI is an active optical cable (about 5 m), "source" end at the PC.
- **Power:** a new double socket low on the side wall behind the sofa (electrician) takes the PC and the rear speakers, so the PC's 500–850 W stays off the media wall's strips.
- **Keyboard and mouse:** no tray over the sofa; a wireless set on a lap board lives on the shelf or in the plinth drawer.
- **Headroom:** the shelf's underside is at 151.4, and it covers only the 28 cm of the sofa nearest the wall.

7 · The oak joinery

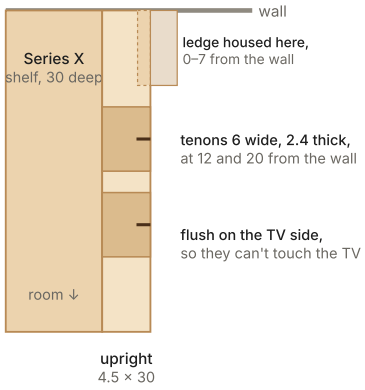
Above the worktop the wall is furniture rather than cladding: two solid oak uprights, shelves carried on joints you can see, and the Card Room Green wall showing between them.

- **Uprights:** solid oak, 4.5 × 30 cm, 159 cm long, each on two 10 mm dowels into the worktop, scribed to the ceiling and held to the back wall with concealed brackets; the left one also screws to the side wall.
- **Wedged through-tenons:** the Series X shelf (top 134) passes through the right upright on two tenons, 6 cm wide and 2.4 thick, 12 and 20 cm from the wall. They're wedged and planed flush: standing proud, they'd touch the TV.
- **Housed and pegged:** the high shelf (two lengths, top 196) and the record ledge (top 155.4) are housed 1.2 cm into the uprights and pinned with 10 mm oak pegs, which show as dark dots.
- **LED grooves:** a routed groove on each upright's inner face holds a 2700K strip that washes the wall either side of the TV.
- **Hidden cables:** a 2 × 1.5 cm groove down the back of the right upright carries Deco B's 12 V lead and patch lead, the Series X's HDMI and power, and the vinyl lead.
- **Finish:** hardwax oil throughout; lightly limed or smoked oak avoids the bright orange "new oak" look.

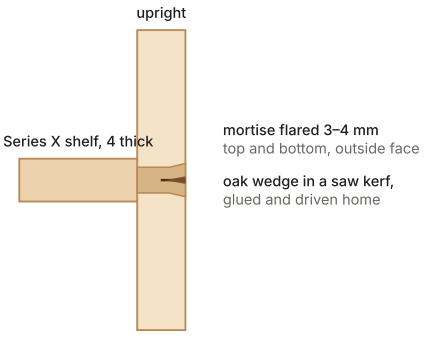
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenons, from above

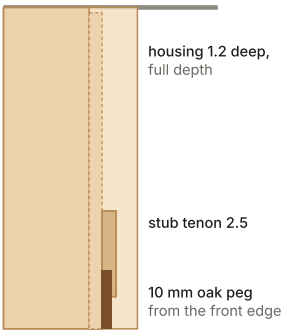


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged



high shelf / ledge
seen from above

• Through-tenons (A, B): the Series X shelf (top 134) passes through the right upright on two tenons, 12 and 20 cm from the wall, wedged and planed flush: proud, they'd touch the TV

• Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.

• Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets. The left one also screws to the side wall, and

Joinery cut list

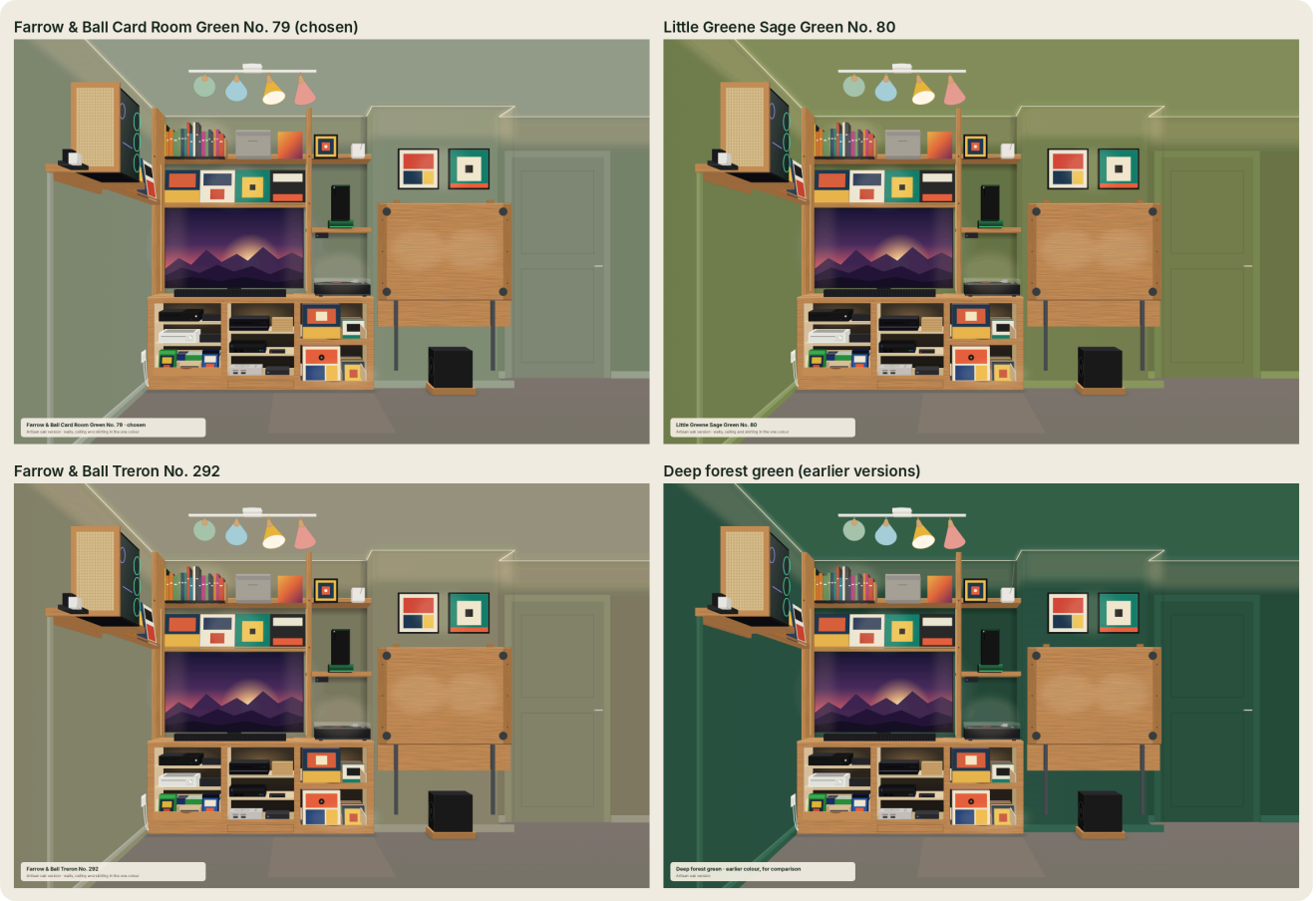
Part	Material	Finished size (cm)	Joints	Qty
Uprights	Solid oak, quarter-sawn if you can	159 × 30 × 4.5 (buy 165 and scribe)	Housings for the high shelf and ledge; mortises for the Series X shelf in the right one; LED groove inside; cable groove at the back of the right one	2
High shelf	Solid oak	128.9 × 30 × 4 and 51.7 × 30 × 4	Housed 1.2 into the uprights, pegged; the short length on a cleat on the reveal	2

Record ledge · wrap ledge	Solid oak	128.9 × 7 × 2.4 · 134 × 7 × 2.4; lips 0.9 × 1.5	Housed and pegged · cleat and two dowels	1 · 1
Series X shelf	Solid oak	55 × 30 × 4 (50.5 + 4.5 through)	Two through-tenons 6 × 2.4, wedged flush; cleat on the reveal	1
PC shelf	Solid oak	82 × 28 × 4	Oak cleat and two concealed brackets	1
Worktop and backboard	One solid oak worktop, 300 × 62	186 × 46 × 3 and 110 × 22 × 3	Chamfered fronts; cut edges sealed	1 board
Face frame	Oak, 20 mm	186 × 73, stiles about 5 wide	Glued and pinned; end stiles scribed	1 set
Wedges · pegs · dowels	Oak · 10 mm oak dowel	6 × 3, tapering · 5 long	Glued	2 + spares · 6 · 8
PC screen	Oak frame 2 × 3, pre-woven cane	26 × 48	Cane in a groove with a spline	1

8 · Paint and light

Paint: Farrow & Ball Card Room Green No. 79

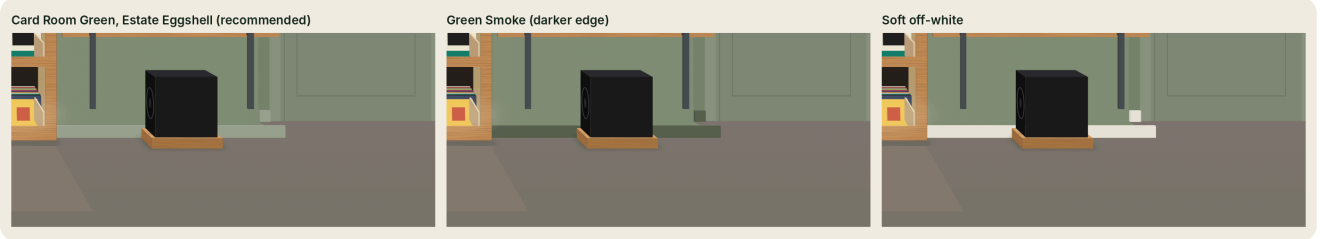
The room is colour-drenched in Card Room Green: walls, ceiling, alcove, chimney breast, skirting, door and frame, radiator and the trunking. It's a soft grey-green that lets the oak and the records lead.



The v11 wall in Card Room Green (chosen) beside Sage Green, Treron and the earlier deep green. Screen colours are approximate and the renders are lit as in the evening: judge from painted samples.

- **Finishes:** Estate or Modern Emulsion on walls, ceiling, alcove and breast; Estate Eggshell in the same colour on skirting, door, frame, radiator and trunking (Modern Eggshell if the skirting takes knocks).
- **Order:** paint before anything is fixed, so nothing needs cutting in round the uprights, shelves or the dining wall.

Skirting: Card Room Green too



Recommended: Card Room Green in Estate Eggshell, which completes the drench and hides scuffs far better than white. Darker option: Green Smoke. Over the white gloss: wash, key, woodwork primer, two coats of eggshell.

Light, in three warm layers

- **The LED strip on the ceiling** (as now): round the room's edge, into the alcove and across the breast; 2700K on a dimmer.
- **Built into the oak:** a 2700K strip in each upright's groove, on a driver on the fused spur; a 6500K bias light on the back of the TV from its USB port.
- **Two spot bars** in the style of Dunelm's Wolston (£49 each; 84 × 16 cm, 22.4 cm drop; 4 × E14, 6 W max[14]), parallel to the media wall, on your two existing ceiling points. Bar 1 hangs on the rose straight out from the TV unit, halfway to the sofa. Bar 2 hangs on the point between the door and the west wall, and its heads swing to the table. Each gets an LED-compatible dimmer, so no new wiring is needed. Bar 1 sits in front of the screen, so tilt its heads up and away: none should be visible in the screen from the sofa.

Head	Bar 1, on the existing rose	Bar 2, over the dining area
Sage	The books and the left of the high shelf	The framed LPs above the table
Pale blue	The record ledge above the TV	The table (the folded top, when it's up)
Mustard	Down onto the coffee table	The table
Dusty pink	The turntable and the Series X	Down onto the way in, by the door

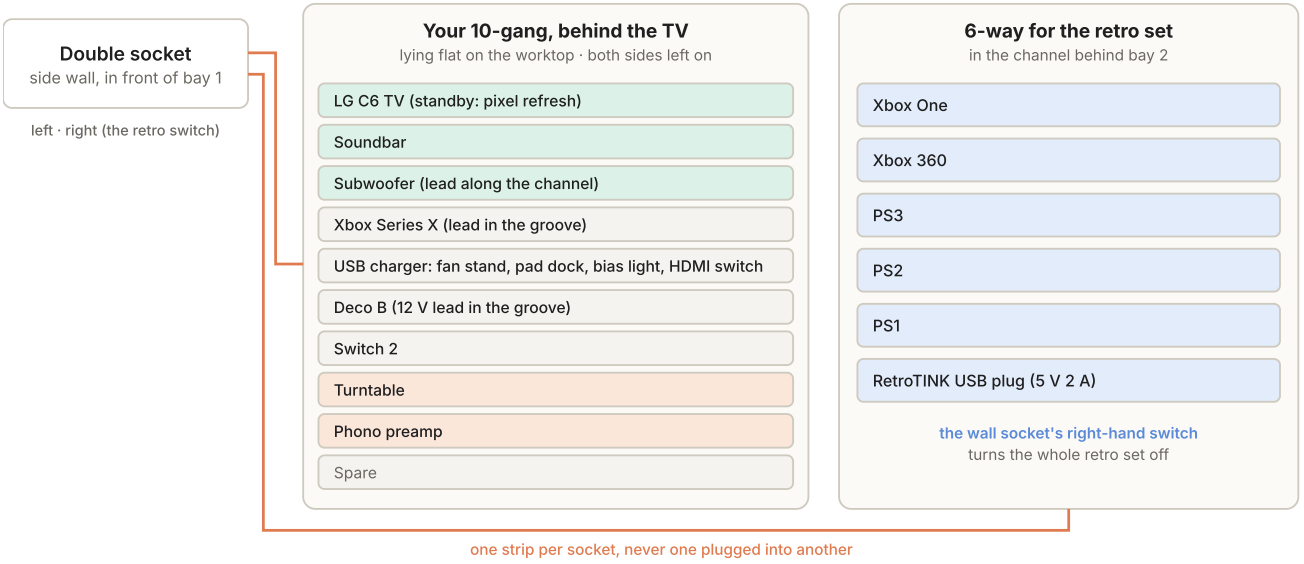
Never point a head at the TV. For six or eight the table stands right under bar 2: point its blue and mustard heads straight down onto it. Use warm-white (2700K) dimmable E14 LED bulbs throughout.

9 · Power, cables and network

Power

Power map: your 10-gang plus a 6-way for the retro set

Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



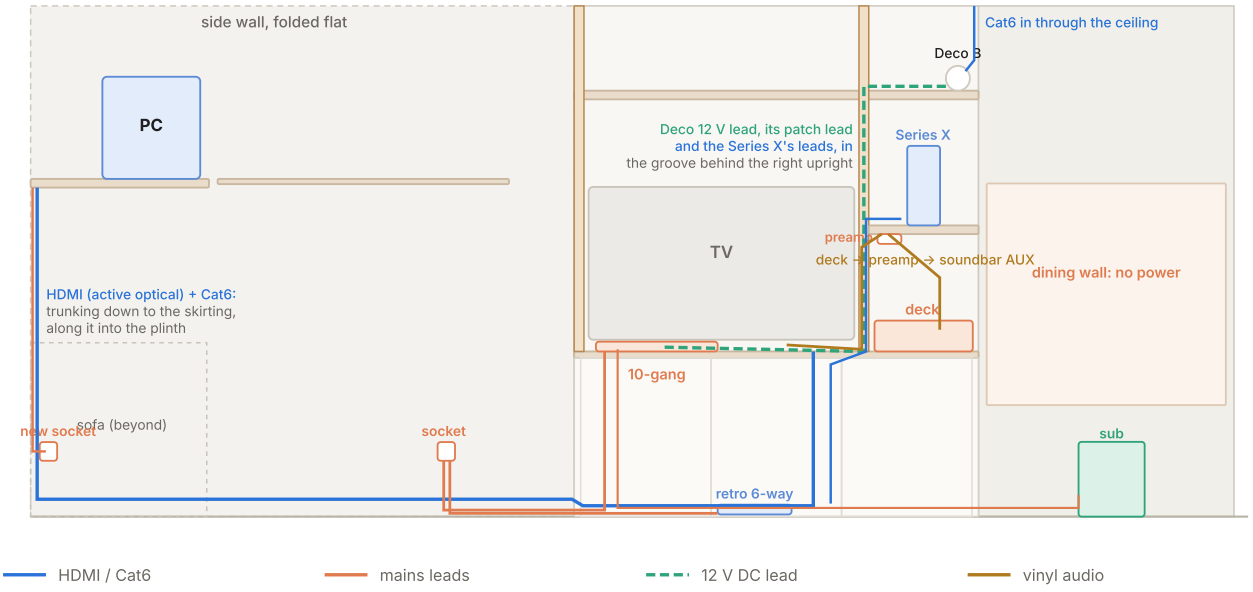
- The socket on the side wall is a single today: the electrician swaps it for a double, at the same spot.
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- PC and rear speakers: a new double socket low on the side wall behind the sofa; the PC (500–850 W) stays off these strips.
- The sub plugs into the 10-gang: its lead runs along the channel and the breast's skirting. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 0.7 kW with everything on, the 6-way up to about 0.8 kW with a launch PS3 running; each well under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

- **The socket:** the single on the side wall just in front of bay 1, about 30 cm up, becomes a double at the same spot. The left outlet feeds the 10-gang; the right one feeds the retro 6-way, so its own switch is now the retro switch. The 10-gang's switches sit behind the TV, 2 cm from the corner post, where you can't reach them any more.
- **10-gang:** lying flat on the worktop behind the TV, both sides left on: TV, soundbar, sub, Series X, USB charger, Deco B, Switch 2, the deck and the preamp.
- **Retro 6-way:** in the channel behind bay 2: Xbox One, Xbox 360, PS3, PS2, PS1, RetroTINK.
- **OLED care:** leave the TV in standby so it runs its pixel refresh.

Cable routes

Cable routes: everything ends in the channel behind the base

Front view, schematic, with the side wall folded flat to the left so the PC's route shows; the sub sits under the dining wall

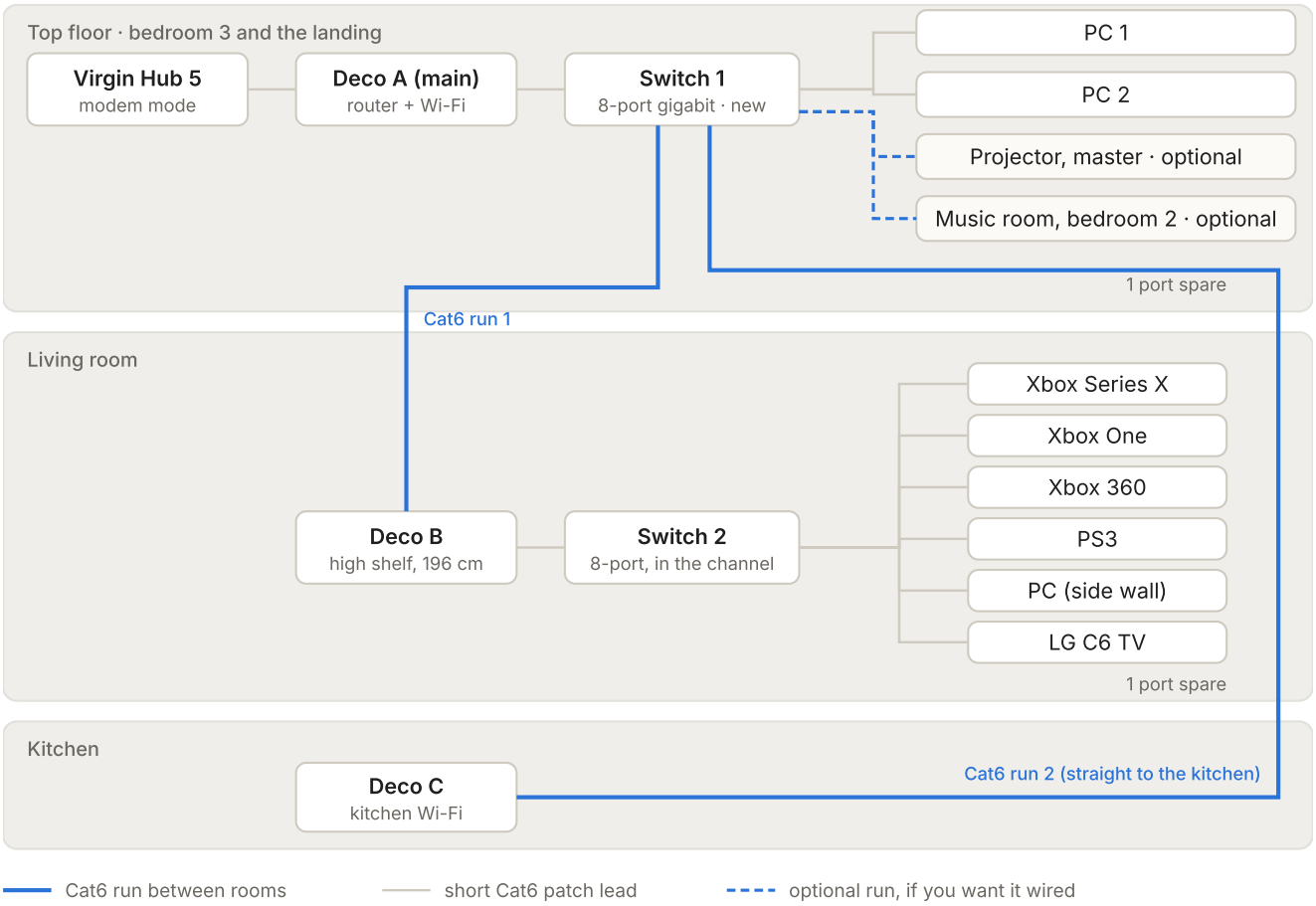


- **The channel:** 4 cm behind the frames, flat two-compartment trunking: mains on one side, HDMI, Cat6 and audio on the other. Switch 2 and the retro 6-way stand on edge in it.
- **The groove behind the right upright:** Deco B's 12 V and patch leads, the Series X's HDMI and power, and the vinyl lead, down to the worktop's slot.
- **The PC:** down the trunking at the far end of its shelf, along the skirting behind the sofa and into the plinth notch.
- **The sub:** its mains lead from the 10-gang runs down the slot, along the channel to the end of bay 3, out at the reveal and along the breast's skirting to the sub, about 70 cm.
- **Airflow:** console bays stay open-backed; the Series X has 15 cm each side and 21 above.

Network

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



- **Bedroom 3:** the Hub 5 (in modem mode), Deco A and Switch 1 sit together with the gaming PCs. Switch 1's eight ports: the uplink, two PCs, the two Cat6 runs down to Deco B and Deco C, and, if you want them, the projector in the master and the music room in bedroom 2.
- **The route down:** across the landing to the airing cupboard, which sits over the alcove's corner, and down through its floor. Run 1 comes through the living room ceiling in the alcove's right corner to Deco B; run 2 follows the right upright's groove down and through the floor to the kitchen, where Deco C goes on the outside of the brick cupboard, up high.
- **Sloped ceilings:** bedroom 2 and the bathroom slope down to the back wall; keep any Deco or desk in bedroom 2 on the high side.

Stage 1, mesh up today: unplug everything from the hub, log in at 192.168.0.1 and turn on Modem Mode (its page then moves to 192.168.100.1); set up Deco A with the old Wi-Fi name and password; Switch 1 into Deco A's other port and the PCs into Switch 1; add Deco B and Deco C over Wi-Fi for now; check the Xbox shows NAT: Open.

Stage 2, after the cables: plug each Cat6 run into its Deco, confirm Deco B and C show as "Ethernet" in the app, and give the Series X and PC high priority under QoS.

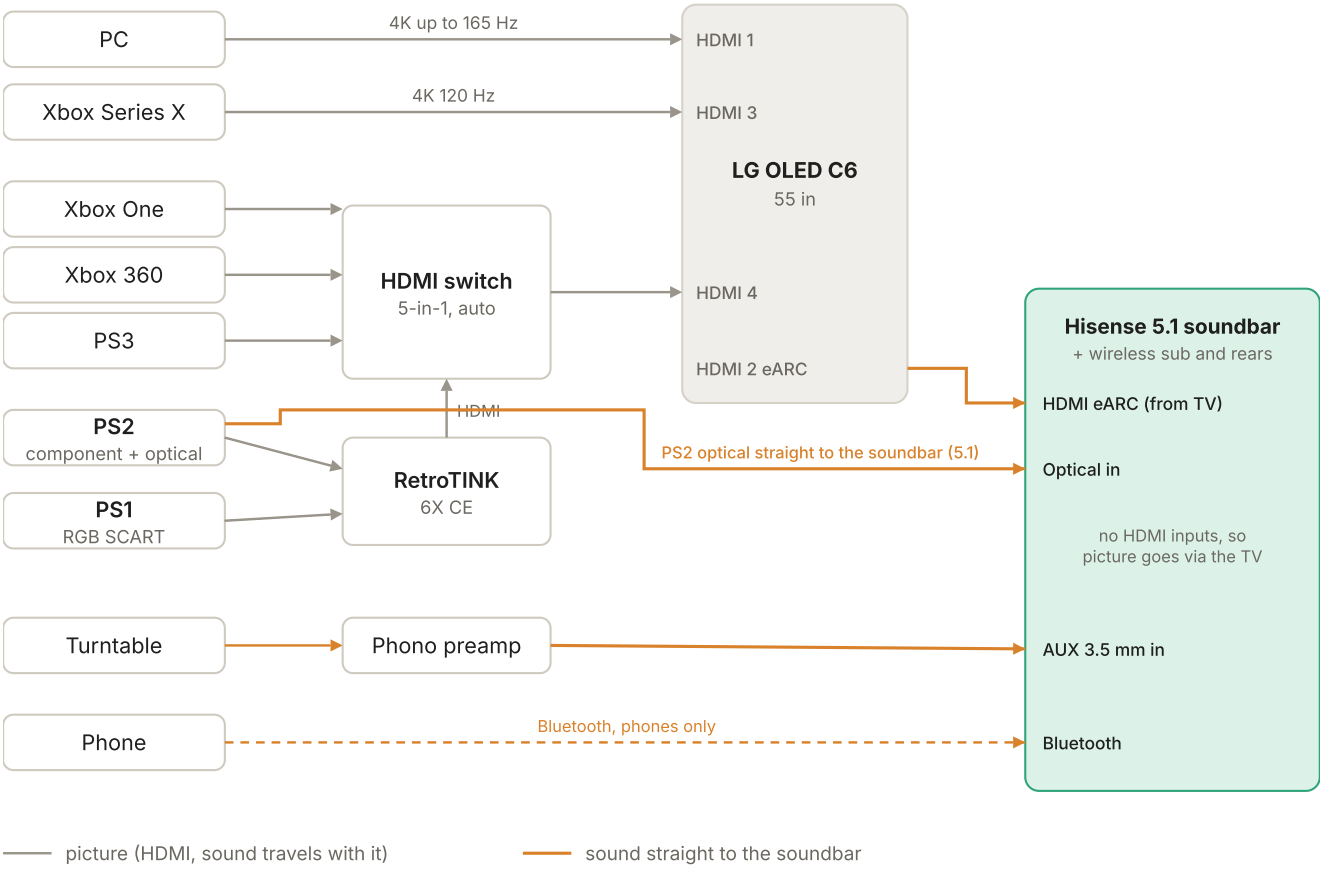
Cabling rules

- Solid pure-copper Cat6, round; not CCA or flat cable for the long runs. Measure the real route and add 2–3 m.
- Cross mains at right angles; never staple through a cable; no kinks; label both ends. Never two cables between the same two devices.
- **Done when:** Deco B and C show "Ethernet", each PC shows 1.0 Gbps, the Xbox shows NAT: Open, and phones hold a strong signal in the kitchen and across the living room.

AV: everything sounds through the Hisense

Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



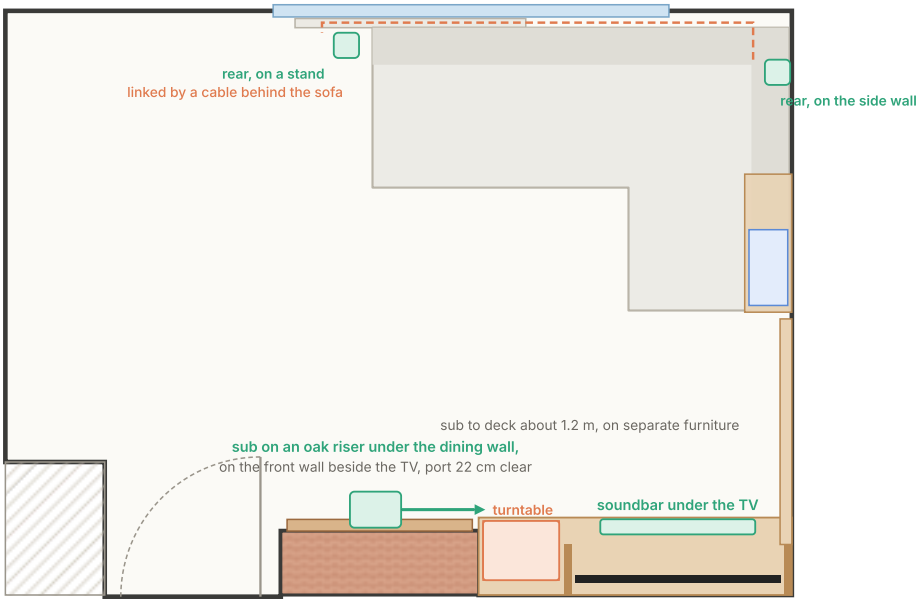
Source	Picture	Sound
PC · Xbox Series X	TV HDMI 1 · HDMI 3	Via the TV over eARC (5.1 or Atmos)
Xbox One, Xbox 360, PS3	HDMI switch → TV HDMI 4	Via the TV over eARC, 5.1
PS2	Component → RetroTINK → switch → TV	Optical straight to the soundbar
PS1	RGB SCART → RetroTINK → switch → TV	Stereo through the RetroTINK and eARC
Turntable · phone	—	Preamp → AUX · Bluetooth

TV settings: rename the inputs (HDMI 4 "Retro"), Deep Colour on HDMI 1, 3 and 4, Game Optimiser on the console inputs, sound out to HDMI eARC. RetroTINK-6X CE: a 5 V 2 A USB plug on the retro 6-way; the PS2's Component Video Out set to Y Cb/Pb Cr/Pr; RGB SCART for the PS1.

Sound layout

Sound: the sub on the front wall with the TV, the rears behind the sofa

To scale, north at the top. The sub is wireless: on the front wall it blends with the soundbar, out of the way of feet



- **Sub:** wireless, on an oak riser under the dining wall, between the folded bracket arms, on its side to the breast with the driver towards the TV and the rear port 22 cm clear. On the front wall, beside the soundbar, its bass blends with the bar instead of coming from behind you; it's out of the walkway, so nobody trips on it or its lead; and the folded table and the oak riser frame it. If the bass booms, turn it down rather than moving it.
- **Rears:** one on the side wall at seated ear height behind the sofa's corner, one on a stand past the sofa's west end; linked by a cable behind the sofa, powered from the new socket.

10 · Build guide, for a first-time builder

This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Where a step is a job for a professional, it says so. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **Mains wiring is the electrician's.** Swapping the socket, the new socket behind the sofa and the dimmers are all theirs. Adding to existing circuits like this isn't normally notifiable under Part P, but a registered electrician certifies it either way.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15
Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			

Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.
2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	**Measure**	The alcove's width at six heights, the floor's level, the breast's flatness, the door's swing, the sofa's size; count your records and games.	Numbers within 1 cm of the plan, or tell me and I'll redraw.	2 h

2	**Network, stage 1**	Hub in modem mode; Decos and switches over Wi-Fi.	Internet in every room.	2 h
3	**Electrician**	Cat6 runs, the double socket, the socket behind the sofa, the second ceiling point and dimmer, the fused spur. Mark every cable route on the wall.	The certificate, and photos of where the cables run.	1–2 days (theirs)
4	**Paint**	The whole room in Card Room Green; skirting and door in eggshell. Let it cure a week before fixing anything.	Two even coats; no white showing at the edges.	2–3 days
5	**Mark the heights**	With the laser: 73 and 76 (worktop), 130 and 134 (Series X shelf), 153 and 155.4 (ledges), 192 and 196 (high shelf); the uprights at 0–4.5 and 131–135.5 from the side wall; the dining wall's backboard at 51.3–73.3, centred on the breast.	Lines level end to end; the uprights' lines plumb.	1 h
6	**Base**	Build the plinth (with the drawer bay under bay 2), level it with shims, stand the two BESTÅ frames on it; build bay 3 from ply and bolt it on. Spacers to the wall, anti-tip brackets, trunking in the channel, cables run in.	The tops level within 1 mm across 186; nothing rocks.	1 day
7	**Worktop**	Scribe the worktop to the walls, cut the TV slot and the grommet hole, oil all over, screw it up through the tops in slotted holes.	Tight to the walls; the slot over the channel.	½ day
8	**Inside the base**	Shelves at the fit-proof heights; the game drawer; the record trays' runners at 2.3 and 37.0, then the trays; the plinth drawer.	Every drawer runs smoothly; trays clear each other by 2 cm.	1–2 days
9	**Uprights and shelves**	Cut the housings and mortises on the bench. Dry-fit each upright with its shelves. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the brackets, then fit the high shelf, the ledge, the Series X shelf (wedge the tenons) and the pegs.	Uprights plumb; shelves level; the TV slides between with 2 cm each side.	2–3 days
10	**Side wall**	The oak cleat for the wrap ledge and the PC shelf; dowel the wrap ledge to the left upright's front edge; the concealed PC brackets; the cane screen; the trunking down to the skirting.	One level line with the ledge over the TV; the PC shelf takes your weight leaning on it.	1 day
11	**Dining wall**	Screw the backboard to the breast (plugs every 20 cm). Fit the two Hebgo brackets on its face, 80 apart, arm tops level with its top. Screw the lift-off hinges to its top edge and the table's back edge. Hang the table, fit the turn buttons at the top, fit the leg plates underneath.	Folds up and down without binding; brackets lock both sides; it doesn't move when you lean on it.	1 day
12	**Lighting**	LED in the upright grooves, the ceiling strip through its notches, the bias light; aim both spot bars.	No head shines at the screen or into your eyes from the sofa.	½ day
13	**Connect**	TV, soundbar, consoles, deck, PC; network stage 2; label every plug and input.	The "done when" list in the network section.	1 day

Roughly 12–15 working days spread over a few weekends, plus the electrician and the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the trays before the anti-tip brackets go in.** A full tray weighs about 32 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

11 · Buy list and tracker

Every purchase and job, in build order, comes to about £2,532–4,772 plus the electrician's quote. Estimates are approximate UK prices; changed and new items are marked. Tick items off and write in what you paid.

Phase	Estimate (£)	Biggest item
1 Network, now	51–70	two 8-port switches
2 Electrician and cabling	125–230, plus the electrician	Cat6 and trunking
3 Paint	220–540	paint
4 Base unit	387–747	record trays and the BESTÅ frames
5 Oak joinery	710–1,310	the solid oak worktop
6 Dining wall	511–935	the oak table top
7 Lighting	160–330	two spot bars
8 TV, consoles and AV	288–420	RetroTINK-6X CE
9 Vinyl kit	80–190	phono preamp
10 Finishing	—	—
Total	**2,532–4,772, plus the electrician**	

Already owned: Virgin Hub 5, three Deco X1500s, two TL-SG105S switches (both become spares), LG OLED C6 55" and its stand, a 10-gang, the Hisense AX5100G 5.1 soundbar, the Series X fan stand, the turntable, the L-sofa and all the consoles.

Status	Item	Qty	Estimate (£)	Paid (£)
1 NETWORK, NOW				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
2 ELECTRICIAN AND CABLING				
☐ To do	Book one electrician visit: Cat6 from bedroom 3 down through the airing cupboard to the alcove's right corner (Deco B) and on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2; swap the side-wall socket for a double **at the same spot**; a new double socket low on the side wall behind the sofa for the PC and the rear speakers **(new)**; LED-compatible dimmers for both ceiling points; a fused spur for the LED drivers; fit both spot bars, on the existing rose between the TV and the sofa and the existing point by the door **(changed: no new point)**		Quote	
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To buy	Slim paintable trunking: the PC drop from the shelf to the skirting and along it to the base, about 4 m **(changed: longer)**	1	20–35	
☐ To buy	PC tidy kit: braided sleeve, right-angle C13 lead, shelf-notch grommet	1	10–20	
3 PAINT				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door, frame and trunking; primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
4 BASE UNIT				
☑ Done	Check the alcove back wall for fixings (solid)			
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☐ To do	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid, or plasterboard on studs **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6, 12 mm ply box 52 × 34 × 5, 300 mm push-to-open runners **(moved from bay 3)**	1	30–50	
☐ To buy	Bay 3 carcass for the records: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts **(changed: holds the trays)**	1	45–80	
☐ To buy	Two face-out record tray kits: 12 and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	2	150–240	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
5 OAK JOINERY				
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, wrap ledge 134 × 7 × 2.4, PC shelf 82 × 28 × 4, lips **(changed)**	1 set	200–340	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Cane screen for the PC: oak frame and pre-woven cane, 26 × 48 cm	1	20–40	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
6 DINING WALL				
☐ To buy	Solid oak table top 110 × 80 × 2.7 with pegged breadboard ends, and a 60 × 80 × 2.7 leaf (edge-glued staves from a timber merchant, who can also radius the corners) **(new)**	1 set	220–400	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	
☐ To buy	Eight legs on screw-in plates: tapered oak legs on M8 hanger bolts (to match), or Emuca Ø60 × 710 mm steel legs (cheaper); four for the top, four for the leaf **(new)**	8	70–160	
☐ To buy	Two oak turn buttons, two leaf-alignment dowels and two table clips for the leaf, felt pads **(new)**	1	10–20	
☐ To buy	Four folding chairs, IKEA TERJE or similar (skip if yours fit: seat about 45–48) **(new)**	4	100–160	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Oak riser for the sub, 34 × 25 × 5 cm, with felt or sorbothane feet (from the worktop offcut) **(changed: under the dining wall)**	1	5–15	
☐ To do	Fix the backboard to the breast, hang the brackets and the top, check it folds and locks; set the legs to the right height			
7 LIGHTING				
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, driver; 6500K bias light for the TV	1	40–90	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars (£49 each), plus eight warm-white (2700K) dimmable E14 LED bulbs, 6 W max **(changed: two)**	2	120–160	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
8 TV, CONSOLES AND AV				
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
9 VINYL KIT				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
10 FINISHING				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			

12 · Still to confirm, later ideas, sources

Still to confirm

- ☐ The L-sofa's size (the plan uses 247 × 168 × 80 high) and which way its return points.
- ☐ The door to the hall: which side it hinges, and whether it opens into the room (the dining chairs on that side sit in its swing).
- ☐ The west end's sizes: which of 935, 1045, 660 and 820 are the recess, the door and the stair box.
- ☐ The turntable's model (the zone takes up to 45.2 × 35.2 cm) and whether it has a PHONO/LINE switch.
- ☐ The PC case's model (the shelf allows 23 × 45 × 47 tall) and that its window is on the left.
- ☐ The Series X fan stand's model: the shelf takes a footprint up to about 30 cm wide.
- ☐ Dining chairs: use ones you have (seat 45–48), or four folding ones? And where the leaf and the legs live (the garage?).
- ☐ The side wall and the chimney breast: solid, or plasterboard? The ledge, PC shelf and dining wall all fix to them.
- ☐ The 23 × 22 cm box in the window's west corner.
- ☐ Records: roughly how many LPs and singles (the wall holds about 188 and 240), and are most in thick outer sleeves?
- ☐ Wired ports for the projector (master) and the music room (bedroom 2)?
- ☑ ~~The TV 25.25 cm nearer the side wall, the turntable in the alcove, the Series X above it, the records below.~~
- ☑ ~~A fold-down oak dining table on the chimney breast; the spot bars on the existing ceiling points; the sub under the dining wall.~~
- ☐ The second ceiling point's exact spot (the plan puts it about 35 cm in from the west wall, 2.3 m from the window): measure it from the wall and the window.
- ☑ ~~The ledge wrapping round onto the side wall, with the PC above the sofa.~~
- ☑ ~~Artisan oak, Card Room Green with matching eggshell skirting, every record face-out, the game drawer as designed.~~

Later ideas

- **Dave and Emma's watch log:** a shared film and TV log, like a Spotify Blend for what you watch: log each film or episode, mark whether you watched it together, rate it, and later get suggestions for what to watch next. To be shaped up separately.

Sources

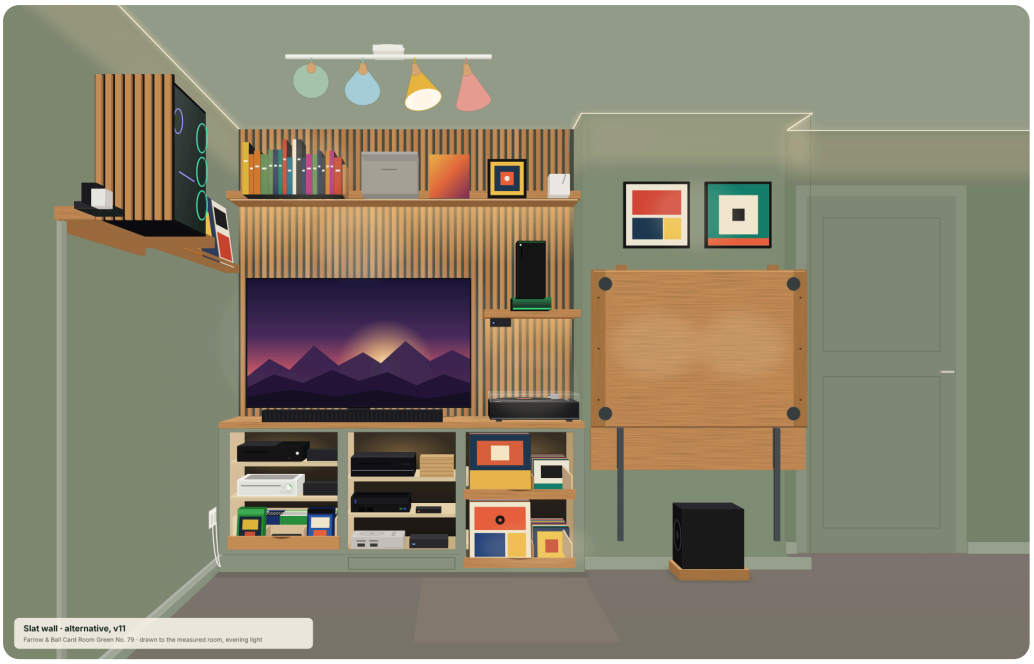
Numbered sources are the item sizes used in the fit proof (section 3).

1. Your hand-measured sketch of the living room, in mm (alcove 1860 × 380, chimney breast 1175, ceiling 2350)
2. LG: OLED55C6 specifications: 1222 × 703 × 45.1 mm without stand, 1222 × 757 × 230 mm with stand, stand 470 × 230 mm, 14.1/15.9 kg, VESA 300 × 200 (lg.com, OLED55C6 product pages)
3. Trusted Reviews and Expert Reviews, Hisense AX5100G reviews: soundbar 920 × 60 × 90 mm, subwoofer 214 × 294 × 304 mm (W × H × D) with a rear-firing port, rear speakers 90 × 144 × 109 mm
4. Wikipedia, Xbox Series X: 151 × 151 × 301 mm; Pure Xbox (Nov 2020) quoting Microsoft: at least 4 cm clear above the console on an open stand, 10–15 cm recommended on all sides
5. IKEA: BESTÅ frame 60 × 40 × 64 cm (10247379); BESTÅ shelf 56 × 36 cm, 20 kg maximum load
6. TP-Link: TL-SG108 specifications, 158 × 101 × 25 mm (tp-link.com)
7. Laptop Outlet: TP-Link Deco X1500 3-pack listing, 110 × 110 × 114 mm per unit
8. Audio-Technica: AT-LP120XUSB specifications, 452 × 352 × 141.6 mm, 8 kg without dust cover (audio-technica.com)

9. RetroTINK: RetroTINK-6X CE shop page (power 5 V 2 A by USB; no size published)
10. UGREEN: 5-in-1 4K60 HDMI switch CM568 (90512), 127 × 59 × 25 mm
11. Original Xbox One power supply: 17 × 7.5 × 4.8 cm (6.69 × 2.96 × 1.88 in), as listed by several replacement-PSU sellers
12. Wikipedia: Comparison of PlayStation consoles (PS1 270 × 188 × 60, PS2 302 × 183 × 78, PS3 325 × 274 × 98 mm); Xbox One (333 × 274 × 79 mm); Neowin: official Xbox 360 dimensions (309 × 258 × 83 mm)
13. Record and case sizes: LP sleeve 31.4 cm (12⅜ in); 7-inch sleeve 18.3–18.4 cm; DVD-style keep case 19 × 13.5 × 1.4 cm; Xbox One case 17 × 13.5 × 1.1–1.2 cm (videogamecasesize.info; Walvis Products; Wikipedia: Keep case)
14. Dunelm: Elements Wolston 4-light spotlight bar, £49: 84 × 16 cm, 22.4 cm drop, shades 15 cm, E14 6 W max (dunelm.com, product 1000189449)
15. Emuca heavy-duty full-extension runners, 400 mm, 68 kg per pair; Accuride DZ3832EC 350 mm, 43 kg; runner fitting allowance 12.5 mm a side
16. Xbox 360 original 203 W power supply: no manufacturer size published; allowance of 21 × 10 × 6 cm used here (measure yours)
17. Turntable Lab: Audio-Technica replacement dust cover for the AT-LP120XUSB (705-140XP-1815), 17.375 × 13.625 × 2.75 in (44.1 × 34.6 × 7.0 cm)
18. Häfele UK, Technology catalogue 2023, p. 7.206: Hebgo heavy-duty folding bracket 287.43.428, arm 580 mm, height 220 mm, 500 kg per pair, locks automatically when raised; up to 800 mm between brackets (Häfele 287.43.419 listing)
19. Screwfix: Union PowerLoad lift-off hinges, 100 × 88 mm, 3-pack (5938T), 120 kg maximum door weight
20. Emuca: table legs Ø60 mm, 710 mm, height adjustable 710–730 mm, 50 kg per leg, screw-on mounting plate (set of 4)
21. IKEA: TERJE folding chair, 43 W × 49 D × 78 H cm, seat height 48 cm, tested for 110 kg
22. Book a Builder UK, small dining room ideas: 600 mm of table edge per diner; table edge to a wall 600 mm tight minimum, 750 mm to aim for, 1000–1100 mm to walk behind; dining height about 750 mm
23. M&S: Easy Velvet left corner sofa, 247 W × 168 D × 80 H cm: a typical corner sofa, used until yours is measured
24. Virgin Media: Hub 5 modem mode
25. RetroTINK: Introducing the RetroTINK-6X CE (inputs including TOSLINK, deinterlacing); RetroTINK-2X Pro listed as legacy; RetroTINK-4K CE
26. Expert Reviews: LG C6 OLED review (4 × HDMI 2.1, one eARC, ports on the left); JB Hi-Fi: LG C6 specifications (165 Hz mode for PC sources)
27. Hisense AX5100G spec sheet (inputs, auto power-on); JB Hi-Fi Business: AX5100G (rears linked by cable, wireless to the soundbar)
28. TP-Link: Deco X1500 specifications (two gigabit ports per unit)
29. Glorious Vinyl Vault specifications (80 LPs and 24.5 kg per drawer), a check on the tray capacity
30. Screwfix: Essentials 964VN 350 mm runners; Metrol: Accuride DZ3832EC; IKEA: BESTÅ drawer runners
31. NIST SP 500-252 / CLIR pub121: Care and Handling of CDs and DVDs (store upright, book style)
32. Sony PS3 and Microsoft Xbox 360 S manuals: ventilation; IKEA UK: secure it
33. Ideal Home: colour drenching; The Paint Shed: colour drenching with Farrow & Ball; Livingetc: sage green shades (Card Room Green, Treron, Sage Green, Green Smoke)
34. Pontata: Card Room Green vs Little Greene Sage Green (approximate screen colours)
35. Homebase and NC Tiles: Farrow & Ball Card Room Green No. 79 in Modern Eggshell and Estate Eggshell
36. haart Medway: floor plans and listing photos for the house (Portsmouth Close)
37. ConsoleMods: PS2 Component Video Out setting
38. GOV.UK, Approved Document P (electrical safety, dwellings): which electrical work is notifiable

Appendix · The slat wall alternative

Kept for reference, drawn to the v11 layout. The base, the records, the consoles, the side wall and the dining wall are identical. Above the worktop, oak slat panels on battens run from the worktop to the ceiling instead of the uprights; the shelves sit on hidden cleats and the light comes from strips under each shelf. Quicker to fit, but it loses the record ledge over the TV and the joints on show.



Options we rejected along the way

Option	Why not
The vinyl cabinet on the chimney breast (v10)	The breast is the only clear wall for a table; the turntable fits in the alcove now
A keyboard tray under the PC shelf	The shelf is over the sofa; a wireless set on a lap board instead
Hub as router, Decos as access points	Loses HomeShield and parental controls
All-wireless mesh · powerline	Speed drops at each hop · slower and less stable than Cat6
Records spine-up in one drawer	Showed only a handful of covers; face-out trays show every one
A green feature wall only · white skirting	The drench ties the room together; white cuts a bright line round it
Bluetooth from the turntable	Compresses the sound; the AUX lead keeps full quality